

Placeonix LMS

Comprehensive Test Cases Report

Module: LMS Test Cases

Test ID	Module	Test Scenario	Expected Result	Priority	Status	Tested By	Bug ID
AUTH-001	Authentication	Login with valid credentials	User logs in successfully	High			
AUTH-002	Authentication	Login with wrong password	Show invalid credentials	High			
AUTH-003	Authentication	Forgot password	Reset email sent	High			
USER-001	Users	Create Student	Student created	High			
USER-002	Users	Create Mentor	Mentor created	High			
RBAC-001	RBAC	Student opens Admin page	Access denied	High			
COURSE-001	Courses	Create Course	Course created	High			
COURSE-002	Courses	Publish Course	Visible to students	High			
LEARN-001	Learning	Student Enrollment	Enrollment successful	High			
LEARN-002	Learning	Complete Lesson	Progress updated	High			
VIDEO-001	Video	Upload Recording	Upload successful	High			
VIDEO-002	Video	Student Playback	Video streams	High			
IDE-001	IDE	Open IDE	IDE loads	High			
IDE-002	IDE	Run Code	Output displayed	High			
ASSIGN-001	Assignments	Create Assignment	Assignment created	High			
ASSIGN-002	Assignments	Student Submission	Submission stored	High			
QUIZ-001	Quiz	Start Quiz	Quiz starts	High			
QUIZ-002	Quiz	Timer Ends	Auto submit	High			
PAY-001	Payments	Create Order	Order created	High			
PAY-002	Payments	Successful Payment	Course activated	High			
CERT-001	Certificates	Generate Certificate	Certificate generated	High			
PLACE-001	Placement	Create Job	Job listed	High			

Test ID	Module	Test Scenario	Expected Result	Priority	Status	Tested By	Bug ID
ZOHO-001	Zoho	Create Student Email	Email account created	High			
NOTIF-001	Notifications	Send Email	Email delivered	High			
SEC-001	Security	JWT Tampering	Access denied	Critical			
SEC-002	Security	NoSQL Injection	Blocked	Critical			
PERF-001	Performance	500 Concurrent Users	Stable	High			
PERF-002	Performance	1000 Concurrent Users	Stable	High			
ADMIN-001	Admin	Dashboard Statistics	Correct data	High			
EDGE-001	Edge Case	Internet disconnect during upload	Graceful recovery	High			
EDGE-002	Edge Case	Duplicate payment webhook	Single payment recorded	Critical			

Module: Authentication

Test ID	Test Scenario	Expected Result	Priority
AUTH-001	Login with valid email & password	Login successful	High
AUTH-002	Login with invalid password	Show "Invalid Credentials"	High
AUTH-003	Login with invalid email	Show "User Not Found"	High
AUTH-004	Login with empty email	Validation error	High
AUTH-005	Login with empty password	Validation error	High
AUTH-006	Login with both fields empty	Validation error	High
AUTH-007	Login with inactive account	Login denied	High
AUTH-008	Login with blocked account	Login denied	High
AUTH-009	Login with deleted account	Login denied	High
AUTH-010	Login using official Zoho email	Login successful	High
AUTH-011	Login using personal email (if not allowed)	Access denied	Medium
AUTH-012	Login with leading/trailing spaces	Spaces ignored	Medium
AUTH-013	Email is case insensitive	Login successful	Medium
AUTH-014	Password is case sensitive	Login fails for incorrect case	High
AUTH-015	Show loading indicator during login	Loader displayed	Low
AUTH-016	Forgot Password with valid email	Reset email sent	High

Test ID	Test Scenario	Expected Result	Priority
AUTH-017	Forgot Password with invalid email	Error displayed	High
AUTH-018	Forgot Password with empty email	Validation error	High
AUTH-019	Password reset link expired	Show expiration message	High
AUTH-020	Password reset with valid token	Password updated	High
AUTH-021	Password reset with invalid token	Access denied	High
AUTH-022	New password doesn't meet policy	Validation error	High
AUTH-023	Confirm password mismatch	Validation error	High
AUTH-024	Change password after login	Password updated	High
AUTH-025	Login with old password after change	Login fails	High
AUTH-026	Login with new password	Login successful	High
AUTH-027	Logout	Session destroyed	High
AUTH-028	Logout from all devices	All sessions invalidated	High
AUTH-029	Access API after logout	Unauthorized	High
AUTH-030	Session expires automatically	Redirect to login	High
AUTH-031	Refresh Token generates new Access Token	Success	High
AUTH-032	Expired Refresh Token	Login required	High
AUTH-033	Invalid JWT	Unauthorized	High
AUTH-034	Tampered JWT	Unauthorized	Critical
AUTH-035	Multiple device login	Sessions created correctly	Medium
AUTH-036	SQL/NoSQL Injection in email	Blocked	Critical
AUTH-037	SQL/NoSQL Injection in password	Blocked	Critical
AUTH-038	XSS in login fields	Sanitized	Critical
AUTH-039	Brute-force login attempts	Account temporarily locked or rate-limited	Critical
AUTH-040	Too many requests	HTTP 429	High
AUTH-041	Password stored in database	Stored as Argon2/bcrypt hash	Critical
AUTH-042	JWT secret exposed	Never exposed	Critical
AUTH-043	Refresh Token stored securely	Success	Critical
AUTH-044	API without JWT	HTTP 401	High
AUTH-045	Access another user's profile	Access denied	High
AUTH-046	Student admission approved	Zoho email created	High
AUTH-047	Welcome email sent	Email received	High

Test ID	Test Scenario	Expected Result	Priority
AUTH-048	Email already exists	Error handled	Medium
AUTH-049	Disable student account	Zoho account disabled	High
AUTH-050	Reset Zoho password	Password reset	High
AUTH-051	POST /login	HTTP 200	High
AUTH-052	POST /login invalid	HTTP 401	High
AUTH-053	POST /forgot-password	HTTP 200	High
AUTH-054	POST /reset-password	HTTP 200	High
AUTH-055	POST /logout	HTTP 200	High
AUTH-056	POST /refresh-token	HTTP 200	High
AUTH-057	GET /profile	Returns logged-in user	High
AUTH-058	GET /profile without token	HTTP 401	High
AUTH-059	GET /sessions	Returns active sessions	Medium
AUTH-060	DELETE /sessions/:id	Session terminated	Medium
AUTH-061	Refresh browser during login	No duplicate request	Medium
AUTH-062	Double-click Login button	Single login request	Medium
AUTH-063	Internet disconnect during login	Proper error message	High
AUTH-064	Server unavailable	Friendly error message	High
AUTH-065	Browser Back after logout	User remains logged out	High
AUTH-066	Login from two browsers	Both sessions tracked	Medium
AUTH-067	Change password while logged in elsewhere	Old session invalidated (if configured)	Medium
AUTH-068	Token expires during API request	Refresh flow or re-login prompt	High
AUTH-069	Login after account disabled	Access denied	High
AUTH-070	Very long email/password input	Validation handles safely	Medium

Module: User

Test ID	Test Scenario	Expected Result	Priority
USER-001	Create Student	Student account created successfully	High
USER-002	Create Mentor	Mentor account created successfully	High
USER-003	Create Admin	Admin account created successfully	High
USER-004	Create HR	HR account created successfully	Medium
USER-005	Create Recruiter	Recruiter account created successfully	Medium

Test ID	Test Scenario	Expected Result	Priority
USER-006	Create Super Admin	Super Admin account created successfully	High
USER-007	Duplicate email during user creation	Validation error displayed	High
USER-008	Duplicate phone number (if unique)	Validation error displayed	High
USER-009	Required fields missing	Validation error displayed	High
USER-010	Invalid email format	Validation error displayed	High
USER-011	View own profile	Profile displayed correctly	High
USER-012	Update profile	Profile updated successfully	High
USER-013	Update profile picture	Image updated	Medium
USER-014	Remove profile picture	Default avatar shown	Low
USER-015	Update phone number	Saved successfully	Medium
USER-016	Update address	Saved successfully	Low
USER-017	Update language preference	Preference updated	Low
USER-018	Update timezone	Timezone updated	Low
USER-019	Invalid profile data	Validation error	Medium
USER-020	Profile audit log created	Audit entry stored	Medium
USER-021	Assign batch	Student assigned successfully	High
USER-022	Assign course	Course assigned	High
USER-023	Remove course	Course removed	Medium
USER-024	View enrolled courses	Correct courses displayed	High
USER-025	Update placement status	Status updated	Medium
USER-026	Upload resume	Resume uploaded	Medium
USER-027	View learning progress	Progress displayed	High
USER-028	Soft delete student	Student hidden but retained	High
USER-029	Assign mentor to course	Assignment successful	High
USER-030	Assign mentor to batch	Assignment successful	High
USER-031	Remove mentor assignment	Assignment removed	Medium
USER-032	Update mentor specialization	Updated successfully	Medium
USER-033	View assigned students	Students listed correctly	High
USER-034	Update admin role	Role updated	High
USER-035	Disable admin	Login blocked	High
USER-036	Reactivate admin	Login restored	High
USER-037	View admin activity	Activity displayed	Medium

Test ID	Test Scenario	Expected Result	Priority
USER-038	Delete admin (soft delete)	Admin removed safely	High
USER-039	Assign role	Role assigned	High
USER-040	Remove role	Role removed	High
USER-041	Multiple roles assigned	Permissions work correctly	High
USER-042	Change primary role	Dashboard changes correctly	High
USER-043	Role without permissions	Access denied	High
USER-044	Activate user	User can login	High
USER-045	Deactivate user	Login denied	High
USER-046	Suspend user	Access suspended	High
USER-047	Delete user	Soft delete applied	High
USER-048	Restore deleted user	User restored	Medium
USER-049	Search by name	Matching users displayed	Medium
USER-050	Search by email	Matching user displayed	Medium
USER-051	Search by user code	User displayed	High
USER-052	Filter by role	Correct users listed	Medium
USER-053	Filter by status	Correct users listed	Medium
USER-054	Filter by batch	Correct students displayed	Medium
USER-055	Pagination	Records load correctly	Medium
USER-056	Student admission approved	Official Zoho email created	High
USER-057	Disable student	Zoho account disabled	High
USER-058	Reactivate student	Zoho account reactivated	Medium
USER-059	Reset official email password	Password reset	High
USER-060	Welcome email sent	Student receives credentials	High
USER-061	POST /users	User created	High
USER-062	GET /users	User list returned	High
USER-063	GET /users/:id	User details returned	High
USER-064	PUT /users/:id	User updated	High
USER-065	DELETE /users/:id	User soft deleted	High
USER-066	GET /users?role=student	Filter works	Medium
USER-067	GET /users?status=active	Filter works	Medium
USER-068	Student edits another student's profile	Access denied	Critical
USER-069	Mentor edits admin profile	Access denied	Critical

Test ID	Test Scenario	Expected Result	Priority
USER-070	Unauthorized profile API access	HTTP 401	High
USER-071	Invalid user ID	Proper error returned	Medium
USER-072	Mass assignment attempt	Restricted fields ignored	Critical
USER-073	Create user with 255-character name	Handled correctly	Medium
USER-074	Upload unsupported profile image	Validation error	Medium
USER-075	Create 1,000 users via import	Successful without failures	Medium
USER-076	Simultaneous profile updates	No data corruption	High
USER-077	Delete user with active enrollments	Business rules enforced	High
USER-078	Restore soft-deleted user	All relationships remain intact	High
USER-079	Export user list	Correct data exported	Medium
USER-080	Import users with invalid records	Invalid rows rejected, valid rows imported	Medium

Module: RBAC

Test ID	Test Scenario	Expected Result	Priority
RBAC-001	Create Admin role	Role created	High
RBAC-002	Create Mentor role	Role created	High
RBAC-003	Create Student role	Role created	High
RBAC-004	View all roles	Roles displayed	High
RBAC-005	Update role name	Role updated	Medium
RBAC-006	Assign Admin role	Assigned successfully	High
RBAC-007	Assign Mentor role	Assigned successfully	High
RBAC-008	Assign Student role	Assigned successfully	High
RBAC-009	Change user role	Role updated	High
RBAC-010	Remove user role	Role removed	Medium
RBAC-011	Student opens Student Dashboard	Allowed	High
RBAC-012	Student opens Mentor Dashboard	Denied	High
RBAC-013	Student opens Admin Dashboard	Denied	High
RBAC-014	Mentor opens Mentor Dashboard	Allowed	High
RBAC-015	Mentor opens Student Dashboard	Denied	High
RBAC-016	Mentor opens Admin Dashboard	Denied	High
RBAC-017	Admin opens Admin Dashboard	Allowed	High

Test ID	Test Scenario	Expected Result	Priority
RBAC-018	Admin opens Mentor Dashboard	Allowed	High
RBAC-019	Admin opens Student Dashboard	Allowed	High
RBAC-020	Student calls Admin API	HTTP 403	Critical
RBAC-021	Student creates course	HTTP 403	High
RBAC-022	Mentor creates lesson	HTTP 201	High
RBAC-023	Mentor deletes user	HTTP 403	High
RBAC-024	Admin manages users	HTTP 200	High
RBAC-025	API without JWT	HTTP 401	Critical
RBAC-026	API with expired JWT	HTTP 401	Critical
RBAC-027	Student views enrolled courses	Allowed	High
RBAC-028	Student submits assignment	Allowed	High
RBAC-029	Student edits another student's data	Denied	Critical
RBAC-030	Mentor uploads video	Allowed	High
RBAC-031	Mentor creates assignment	Allowed	High
RBAC-032	Mentor edits assigned course	Allowed	High
RBAC-033	Mentor edits unassigned course	Denied	High
RBAC-034	Admin creates course	Allowed	High
RBAC-035	Admin manages batches	Allowed	High
RBAC-036	Admin generates reports	Allowed	High
RBAC-037	JWT role modified manually	Access denied	Critical
RBAC-038	API payload role changed	Ignored by backend	Critical
RBAC-039	Direct URL access to Admin page	Backend returns 403	Critical
RBAC-040	Student accesses another student's profile	Denied	Critical
RBAC-041	Mentor accesses admin APIs	Denied	High
RBAC-042	Role assigned	Audit log created	Medium
RBAC-043	Role changed	Audit log created	Medium
RBAC-044	Unauthorized access attempt	Logged	High
RBAC-045	User has no role	Access denied	High
RBAC-046	Role deleted while user logged in	User logged out or access updated	High
RBAC-047	Invalid role ID	Validation error	Medium
RBAC-048	Two admins update same user's role	No data corruption	High
RBAC-049	Permission cache refresh	Latest permissions applied	Medium

Test ID	Test Scenario	Expected Result	Priority
RBAC-050	Large number of role checks	Performance remains acceptable	Medium

Module: student dashboard

Test ID	Test Scenario	Expected Result	Priority
STU-001	Student logs into dashboard	Dashboard loads successfully	High
STU-002	Dashboard loads after page refresh	Data remains consistent	High
STU-003	Student with no enrolled courses	Empty state displayed	Medium
STU-004	Dashboard loading indicator	Loader shown until data loads	Low
STU-005	Dashboard loads within acceptable time	Dashboard responsive	High
STU-006	View profile details	Correct details displayed	High
STU-007	Update profile	Changes saved successfully	High
STU-008	Upload profile picture	Image updated	Medium
STU-009	Remove profile picture	Default avatar displayed	Low
STU-010	Invalid profile update	Validation message shown	Medium
STU-011	View enrolled courses	All assigned courses displayed	High
STU-012	Open course details	Course page opens	High
STU-013	Continue Learning button	Opens last viewed lesson	High
STU-014	Search course	Matching courses displayed	Medium
STU-015	Filter completed courses	Correct list displayed	Medium
STU-016	Filter ongoing courses	Correct list displayed	Medium
STU-017	Overall progress displayed	Correct percentage shown	High
STU-018	Lesson completion updates progress	Progress updated instantly	High
STU-019	Module completion	Module marked complete	High
STU-020	Course completion	Course marked complete	High
STU-021	Progress after refresh	Progress persists	High
STU-022	Play lesson video	Video plays	High
STU-023	Pause video	Playback paused	Medium
STU-024	Resume video	Continues from last position	High
STU-025	Video quality selection	Quality changes successfully	Medium
STU-026	Fullscreen mode	Video enters fullscreen	Low
STU-027	Unauthorized video access	Access denied	High

Test ID	Test Scenario	Expected Result	Priority
STU-028	View lesson resources	Resources displayed	High
STU-029	Download PDF	File downloaded	Medium
STU-030	Open external link	Opens successfully	Medium
STU-031	Broken resource link	Error message displayed	Medium
STU-032	View assignments	Assignments listed	High
STU-033	Submit assignment	Submission successful	High
STU-034	Upload assignment file	File uploaded	High
STU-035	Late submission	Warning shown	Medium
STU-036	View assignment result	Marks displayed	High
STU-037	Open coding IDE	IDE loads	High
STU-038	Run code	Output displayed	High
STU-039	Save code	Code saved	High
STU-040	Submit coding assignment	Submission stored	High
STU-041	Auto-save	Code saved automatically	Medium
STU-042	Start quiz	Quiz starts	High
STU-043	Submit quiz	Result generated	High
STU-044	Timer expiry	Quiz auto-submits	High
STU-045	View score	Correct score displayed	High
STU-046	View earned certificates	Certificates listed	High
STU-047	Download certificate	PDF downloaded	High
STU-048	Verify QR code	Certificate verified	High
STU-049	View notifications	Notifications displayed	Medium
STU-050	Mark notification as read	Status updated	Medium
STU-051	Announcement received	Announcement displayed	Medium
STU-052	View attendance	Attendance shown correctly	High
STU-053	Attendance percentage	Percentage calculated correctly	Medium
STU-054	View eligible jobs	Jobs displayed	High
STU-055	Apply for job	Application submitted	High
STU-056	View interview schedule	Schedule displayed	Medium
STU-057	View application status	Correct status shown	Medium
STU-058	View payment history	Payments listed	Medium
STU-059	Download invoice	Invoice downloaded	Medium

Test ID	Test Scenario	Expected Result	Priority
STU-060	View fee status	Correct fee status shown	Medium
STU-061	Student accesses another student's dashboard	Access denied	Critical
STU-062	Invalid JWT token	HTTP 401	Critical
STU-063	Expired session	Redirect to login	High
STU-064	Unauthorized API request	HTTP 403	Critical
STU-065	Internet disconnect while watching video	Playback recovers or error shown	High
STU-066	Browser refresh during quiz	State handled correctly	High
STU-067	Multiple tabs open	Session remains consistent	Medium
STU-068	Simultaneous assignment submission	Only one valid submission recorded	High
STU-069	Large number of enrolled courses	Dashboard performs well	Medium
STU-070	Dashboard opened on mobile	Responsive layout displayed	High

Module: Mentor

Test ID	Test Scenario	Expected Result	Priority
MEN-001	Mentor login	Dashboard loads successfully	High
MEN-002	Dashboard refresh	Latest data displayed	High
MEN-003	Dashboard statistics	Correct values displayed	High
MEN-004	Empty dashboard	Proper empty state shown	Medium
MEN-005	Dashboard responsive on mobile	UI adapts correctly	Medium
MEN-006	View profile	Profile displayed	High
MEN-007	Update profile	Profile updated	High
MEN-008	Upload profile picture	Image updated	Medium
MEN-009	Change password	Password updated	High
MEN-010	Update availability	Availability saved	Medium
MEN-011	View assigned courses	Courses displayed	High
MEN-012	Open course	Course details displayed	High
MEN-013	Edit assigned course	Changes saved	High
MEN-014	Edit unassigned course	Access denied	High
MEN-015	Publish lesson	Lesson visible to students	High
MEN-016	Unpublish lesson	Lesson hidden	Medium
MEN-017	Reorder modules	Order updated	Medium

Test ID	Test Scenario	Expected Result	Priority
MEN-018	Search courses	Correct results	Medium
MEN-019	Create lesson	Lesson created	High
MEN-020	Update lesson	Lesson updated	High
MEN-021	Delete lesson	Lesson removed	High
MEN-022	Preview lesson	Lesson preview works	Medium
MEN-023	Mark lesson as preview	Public preview enabled	Medium
MEN-024	Upload video	Video uploaded successfully	High
MEN-025	Cloudflare processing	Status updated to Ready	High
MEN-026	Replace video	New video available	High
MEN-027	Delete video	Video removed	High
MEN-028	Video thumbnail generated	Thumbnail displayed	Medium
MEN-029	Upload invalid format	Validation error	Medium
MEN-030	Upload PDF	Resource uploaded	High
MEN-031	Upload ZIP	Resource uploaded	Medium
MEN-032	Delete resource	Resource removed	Medium
MEN-033	Download resource	Resource downloaded	Medium
MEN-034	Create assignment	Assignment created	High
MEN-035	Edit assignment	Assignment updated	High
MEN-036	Delete assignment	Assignment deleted	High
MEN-037	Set submission deadline	Deadline saved	Medium
MEN-038	View submissions	All submissions listed	High
MEN-039	Evaluate submission	Marks saved	High
MEN-040	Give feedback	Feedback saved	High
MEN-041	Create quiz	Quiz created	High
MEN-042	Add question	Question saved	High
MEN-043	Edit question	Question updated	High
MEN-044	Delete question	Question removed	Medium
MEN-045	Publish quiz	Students can access	High
MEN-046	View quiz results	Results displayed	High
MEN-047	View assigned students	Student list displayed	High
MEN-048	Search student	Correct student displayed	Medium
MEN-049	View student progress	Progress displayed	High

Test ID	Test Scenario	Expected Result	Priority
MEN-050	View attendance	Attendance displayed	High
MEN-051	Download student report	Report generated	Medium
MEN-052	Mark attendance	Attendance saved	High
MEN-053	Edit attendance	Changes updated	High
MEN-054	View attendance report	Report displayed	Medium
MEN-055	Send announcement	Students receive announcement	High
MEN-056	Send notification	Notification delivered	Medium
MEN-057	Schedule announcement	Announcement sent at scheduled time	Medium
MEN-058	Course completion report	Correct report generated	Medium
MEN-059	Student progress report	Correct report generated	Medium
MEN-060	Assignment report	Correct report generated	Medium
MEN-061	Mentor accesses Admin Dashboard	Access denied	Critical
MEN-062	Mentor edits another mentor's course	Access denied	High
MEN-063	Mentor deletes student account	Access denied	Critical
MEN-064	Invalid JWT	HTTP 401	High
MEN-065	Expired session	Redirect to login	High
MEN-066	Upload 2GB video	Upload handled correctly	High
MEN-067	Internet disconnect during upload	Resume or fail gracefully	High
MEN-068	Two mentors edit same lesson	Conflict handled safely	High
MEN-069	Delete lesson with student progress	Business rule enforced	High
MEN-070	Dashboard with 500+ students	Performance acceptable	High

Module: Admin

Test ID	Test Scenario	Expected Result	Priority
ADM-001	Admin login	Dashboard loads successfully	High
ADM-002	Dashboard statistics	Correct statistics displayed	High
ADM-003	Dashboard refresh	Latest data displayed	High
ADM-004	Dashboard responsive	UI displays correctly	Medium
ADM-005	Empty dashboard	Proper empty state displayed	Medium
ADM-006	View all users	Users displayed	High
ADM-007	Create student	Student created	High

Test ID	Test Scenario	Expected Result	Priority
ADM-008	Create mentor	Mentor created	High
ADM-009	Edit user	User updated	High
ADM-010	Disable user	User login blocked	High
ADM-011	Activate user	User can login	High
ADM-012	Delete user	Soft delete successful	High
ADM-013	Search user	User found	Medium
ADM-014	Filter by role	Correct users displayed	Medium
ADM-015	Export users	Excel generated	Medium
ADM-016	Create course	Course created	High
ADM-017	Edit course	Course updated	High
ADM-018	Publish course	Course visible	High
ADM-019	Archive course	Course archived	Medium
ADM-020	Delete course	Course removed	High
ADM-021	Assign mentor	Mentor assigned	High
ADM-022	Upload thumbnail	Thumbnail updated	Medium
ADM-023	Duplicate course code	Validation error	Medium
ADM-024	Create batch	Batch created	High
ADM-025	Update batch	Batch updated	High
ADM-026	Assign students	Students assigned	High
ADM-027	Assign mentor	Mentor assigned	High
ADM-028	Delete batch	Batch deleted	Medium
ADM-029	Create lesson	Lesson created	High
ADM-030	Update lesson	Lesson updated	High
ADM-031	Delete lesson	Lesson removed	High
ADM-032	Change lesson order	Order updated	Medium
ADM-033	Upload Cloudflare video	Upload successful	High
ADM-034	Replace video	Video updated	High
ADM-035	Delete video	Video removed	High
ADM-036	Video processing	Ready status displayed	Medium
ADM-037	Playback preview	Video plays	Medium
ADM-038	Create assignment	Assignment created	High
ADM-039	Edit assignment	Assignment updated	High

Test ID	Test Scenario	Expected Result	Priority
ADM-040	Delete assignment	Assignment removed	High
ADM-041	View submissions	Submissions displayed	High
ADM-042	Create quiz	Quiz created	High
ADM-043	Add questions	Questions saved	High
ADM-044	Publish quiz	Students can access	High
ADM-045	Delete quiz	Quiz removed	Medium
ADM-046	View coding sessions	Sessions listed	Medium
ADM-047	View code submissions	Correct data displayed	Medium
ADM-048	Reset coding session	Session reset	Medium
ADM-049	View execution logs	Logs displayed	Medium
ADM-050	View attendance	Attendance displayed	High
ADM-051	Edit attendance	Updated successfully	High
ADM-052	Export attendance	Excel generated	Medium
ADM-053	Generate certificate	Certificate generated	High
ADM-054	Reissue certificate	Certificate regenerated	Medium
ADM-055	Verify certificate	Verification successful	High
ADM-056	Add company	Company created	High
ADM-057	Add job	Job created	High
ADM-058	View applications	Applications displayed	High
ADM-059	Schedule interview	Interview created	High
ADM-060	Upload offer letter	Offer uploaded	Medium
ADM-061	View payment history	Payments displayed	High
ADM-062	Verify Razorpay payment	Payment verified	High
ADM-063	Generate invoice	Invoice generated	High
ADM-064	Refund payment	Refund processed	Medium
ADM-065	Create official email	Email created	High
ADM-066	Reset email password	Password reset	High
ADM-067	Disable email	Email disabled	High
ADM-068	Welcome email	Sent successfully	Medium
ADM-069	Send announcement	Delivered to users	High
ADM-070	Send email	Email delivered	High
ADM-071	Send WhatsApp	Message delivered	Medium

Test ID	Test Scenario	Expected Result	Priority
ADM-072	Schedule announcement	Sent on time	Medium
ADM-073	Student report	Correct report	High
ADM-074	Mentor report	Correct report	High
ADM-075	Course report	Correct report	High
ADM-076	Revenue report	Correct report	High
ADM-077	Placement report	Correct report	High
ADM-078	Dashboard analytics	Correct metrics	High
ADM-079	Learning analytics	Accurate data	Medium
ADM-080	Revenue analytics	Accurate totals	Medium
ADM-081	IDE analytics	Correct statistics	Medium
ADM-082	Update organization details	Saved successfully	Medium
ADM-083	Update logo	Logo updated	Medium
ADM-084	Update SMTP settings	Emails sent successfully	High
ADM-085	Update payment settings	Razorpay works	High
ADM-086	Update Cloudflare settings	Video upload works	High
ADM-087	View audit logs	Logs displayed	High
ADM-088	Filter audit logs	Correct results	Medium
ADM-089	Export audit logs	Excel generated	Medium
ADM-090	Invalid JWT	HTTP 401	Critical
ADM-091	Student accesses Admin Dashboard	HTTP 403	Critical
ADM-092	Mentor accesses Admin APIs	HTTP 403	Critical
ADM-093	Unauthorized delete request	Access denied	Critical
ADM-094	Rate limiting	Requests throttled	High
ADM-095	SQL/NoSQL Injection	Blocked	Critical
ADM-096	1000 concurrent dashboard users	System remains stable	High
ADM-097	Delete course with enrolled students	Business rules enforced	High
ADM-098	Delete mentor assigned to batch	Validation prevents deletion	High
ADM-099	Internet disconnect during save	Proper recovery message	Medium
ADM-100	Simultaneous edits by two admins	Conflict handled safely	High

Module: Courses

Test ID	Test Scenario	Expected Result	Priority
COURSE-001	Create a new course with valid details	Course created successfully	High
COURSE-002	Create course without title	Validation error	High
COURSE-003	Create course without category	Validation error	High
COURSE-004	Create duplicate course code	Validation error	High
COURSE-005	Create course with invalid duration	Validation error	Medium
COURSE-006	Upload course thumbnail	Thumbnail uploaded	Medium
COURSE-007	Upload invalid thumbnail format	Validation error	Medium
COURSE-008	Upload banner image	Banner uploaded	Medium
COURSE-009	Update course title	Updated successfully	High
COURSE-010	Update description	Updated successfully	Medium
COURSE-011	Change category	Updated successfully	Medium
COURSE-012	Change mentor	Mentor updated	High
COURSE-013	Update price	Price updated	Medium
COURSE-014	Update discount	Discount updated	Medium
COURSE-015	Update tags	Tags updated	Low
COURSE-016	Publish course	Visible to students	High
COURSE-017	Unpublish course	Hidden from students	High
COURSE-018	Archive course	Course archived	Medium
COURSE-019	Restore archived course	Course restored	Medium
COURSE-020	Delete course (Soft Delete)	Course removed but retained	High
COURSE-021	Assign mentor	Mentor assigned	High
COURSE-022	Assign multiple mentors	All mentors assigned	Medium
COURSE-023	Remove mentor	Mentor removed	Medium
COURSE-024	Mentor views assigned course	Access granted	High
COURSE-025	Mentor edits unassigned course	Access denied	High
COURSE-026	Student enrolls	Enrollment successful	High
COURSE-027	Duplicate enrollment	Prevent duplicate	High
COURSE-028	Cancel enrollment	Enrollment cancelled	Medium
COURSE-029	Student views enrolled course	Course visible	High
COURSE-030	Student opens unpublished course	Access denied	High

Test ID	Test Scenario	Expected Result	Priority
COURSE-031	Search by title	Matching courses displayed	Medium
COURSE-032	Search by mentor	Correct courses displayed	Medium
COURSE-033	Filter by category	Correct list displayed	Medium
COURSE-034	Filter by level	Correct list displayed	Medium
COURSE-035	Sort by newest	Correct order	Low
COURSE-036	Sort by popularity	Correct order	Low
COURSE-037	View course details	Details displayed correctly	High
COURSE-038	View syllabus	Syllabus displayed	High
COURSE-039	View mentor details	Mentor profile displayed	Medium
COURSE-040	View prerequisites	Displayed correctly	Medium
COURSE-041	Modules displayed	Modules visible	High
COURSE-042	Lessons displayed	Lessons visible	High
COURSE-043	Preview lessons	Preview lessons accessible	Medium
COURSE-044	Locked lesson	Access denied	High
COURSE-045	Course resources	Resources visible	Medium
COURSE-046	Student creates course	Access denied	Critical
COURSE-047	Mentor deletes course	Access denied (unless permitted)	High
COURSE-048	Invalid Course ID	Proper error returned	Medium
COURSE-049	Access deleted course	Not found	Medium
COURSE-050	Unauthorized course API	HTTP 401/403	High
COURSE-051	Load 1000 courses	Performance acceptable	Medium
COURSE-052	Search 1000 courses	Results returned quickly	Medium
COURSE-053	Pagination	Correct data loaded	Medium
COURSE-054	Delete course with enrolled students	Business rule enforced	High
COURSE-055	Delete course with assignments	Warning shown	High
COURSE-056	Course without modules	Empty state displayed	Medium
COURSE-057	Course with 500 lessons	Performance acceptable	Medium
COURSE-058	Duplicate course title	Allowed if business rules permit	Low
COURSE-059	Simultaneous course updates	Conflict handled safely	High
COURSE-060	Course thumbnail upload interrupted	Error handled gracefully	Medium
COURSE-061	View enrollment count	Correct count displayed	Medium
COURSE-062	View completion percentage	Correct percentage	Medium

Test ID	Test Scenario	Expected Result	Priority
COURSE-063	View active learners	Correct count	Medium
COURSE-064	View average progress	Correct analytics	Medium
COURSE-065	View course popularity	Correct ranking	Medium
COURSE-066	Course created	Audit log created	Medium
COURSE-067	Course updated	Audit log created	Medium
COURSE-068	Course deleted	Audit log created	High
COURSE-069	Mentor assigned	Audit log created	Medium
COURSE-070	Course published	Audit log created	Medium

Module: Lesson

Test ID	Test Scenario	Expected Result	Priority
LP-001	Open lesson for first time	Progress initialized	High
LP-002	Complete lesson	Lesson marked as completed	High
LP-003	Reopen completed lesson	Completion status retained	High
LP-004	Leave lesson halfway	Progress saved	High
LP-005	Resume lesson	Continues from last position	High
LP-006	Refresh page while learning	Progress retained	High
LP-007	Open next lesson	Next lesson unlocked (if applicable)	High
LP-008	Mark lesson incomplete (Admin/Mentor)	Status updated	Medium
LP-009	Overall course progress	Correct percentage displayed	High
LP-010	Module progress	Correct percentage displayed	High
LP-011	Lesson progress after completion	Updated instantly	High
LP-012	Progress after refresh	Progress persists	High
LP-013	Progress across devices	Synced correctly	High
LP-014	Multiple completed lessons	Overall progress recalculated	High
LP-015	Progress after lesson deletion	Progress recalculated	Medium
LP-016	Complete previous lesson	Next lesson unlocked	High
LP-017	Skip prerequisite lesson	Access denied	High
LP-018	Open locked lesson	Locked message displayed	High
LP-019	Admin unlocks lesson	Student gains access	Medium
LP-020	Mentor unlocks lesson	Based on permissions	Medium

Test ID	Test Scenario	Expected Result	Priority
LP-021	Watch video for 5 minutes	Watch history updated	High
LP-022	Pause video	Last position saved	High
LP-023	Resume video	Starts from last watched position	High
LP-024	Watch complete video	Lesson eligible for completion	High
LP-025	Rewatch completed lesson	Progress unchanged	Medium
LP-026	Bookmark lesson	Bookmark saved	Medium
LP-027	Remove bookmark	Bookmark removed	Medium
LP-028	View bookmarked lessons	Correct list displayed	Medium
LP-029	Bookmark after refresh	Bookmark persists	Medium
LP-030	Video watched completely	Lesson marked complete	High
LP-031	Quiz required before completion	Completion blocked until quiz submitted	High
LP-032	Assignment required	Completion blocked until assignment submitted	High
LP-033	Manual completion by mentor	Status updated	Medium
LP-034	Auto completion	Triggered correctly	High
LP-035	Daily learning time	Correct value displayed	Medium
LP-036	Weekly learning report	Accurate report	Medium
LP-037	Monthly learning report	Accurate report	Medium
LP-038	Learning streak	Correct streak displayed	Medium
LP-039	Most viewed lesson	Analytics updated	Low
LP-040	Continue lesson on another device	Progress synced	High
LP-041	Simultaneous learning on two devices	Progress handled correctly	Medium
LP-042	Logout while watching	Progress saved	High
LP-043	Student accesses another student's progress	Access denied	Critical
LP-044	Modify progress API manually	Validation prevents unauthorized changes	Critical
LP-045	Invalid enrollment ID	Proper error returned	Medium
LP-046	Invalid lesson ID	Proper error returned	Medium
LP-047	Unauthorized progress update	HTTP 403	High
LP-048	Student with 500 completed lessons	Progress loads quickly	Medium
LP-049	Batch progress calculation	Accurate results	Medium
LP-050	Dashboard progress load	Fast response	Medium

Test ID	Test Scenario	Expected Result	Priority
LP-051	Internet disconnect during video	Resume from last saved position	High
LP-052	Browser crash	Progress retained	High
LP-053	Refresh during lesson completion	No duplicate completion	High
LP-054	Lesson removed after completion	Course progress recalculated	Medium
LP-055	Duplicate progress update request	Only one update recorded	High
LP-056	Course archived while learning	Appropriate access/message shown	Medium
LP-057	Progress update fails due to DB issue	Error handled gracefully	High
LP-058	Lesson completed	Audit log created	Medium
LP-059	Progress updated	Audit log created	Medium
LP-060	Lesson unlocked	Audit log created	Medium

Module: Learning

Test ID	Test Scenario	Expected Result	Priority
CL-001	Continue learning from dashboard	Opens last incomplete lesson	High
CL-002	Resume after logout/login	Opens last saved lesson	High
CL-003	Resume after browser refresh	Progress retained	High
CL-004	Resume after system restart	Continues correctly	High
CL-005	Resume on another device	Last position synchronized	High
CL-006	Stop video at 5 minutes	Position saved	High
CL-007	Stop video at 50%	Position saved	High
CL-008	Pause video	Position updated	Medium
CL-009	Close browser without pausing	Position retained	High
CL-010	Video completed	Position reset and lesson marked complete	High
CL-011	Continue Learning card displayed	Latest active course shown	High
CL-012	Multiple active courses	Most recent course shown	Medium
CL-013	No active courses	Card hidden or empty state displayed	Medium
CL-014	Course completed	Card removed	Medium
CL-015	Course archived	Card hidden	Medium
CL-016	Continue after lesson completion	Opens next lesson	High
CL-017	Previous lesson	Opens previous lesson	Medium

Test ID	Test Scenario	Expected Result	Priority
CL-018	Skip locked lesson	Access denied	High
CL-019	Jump to unlocked lesson	Allowed	Medium
CL-020	Last lesson completed	Course marked complete	High
CL-021	Continue on mobile after desktop	Progress synchronized	High
CL-022	Continue on desktop after mobile	Progress synchronized	High
CL-023	Two devices open same lesson	Latest valid progress retained	Medium
CL-024	Simultaneous video watching	No progress corruption	Medium
CL-025	Dashboard updates after lesson completion	Continue card updated	High
CL-026	Dashboard updates after course completion	Continue card removed	High
CL-027	Dashboard shows correct progress	Progress bar accurate	High
CL-028	Dashboard refresh	Continue data persists	High
CL-029	Progress saved in database	Data stored correctly	High
CL-030	Last watched lesson stored	Correct lesson ID	High
CL-031	Last watched timestamp updated	Timestamp accurate	Medium
CL-032	Duplicate progress updates	Single valid record	High
API Testing			
Test ID	API	Expected Result	Priority
CL-033	GET /continue-learning	Returns latest lesson	High
CL-034	GET /continue-learning without token	HTTP 401	High
CL-035	POST /progress/update	Progress saved	High
CL-036	Invalid lesson ID	HTTP 400/404	Medium
CL-037	Student accesses another student's continue data	Access denied	Critical
CL-038	Tampered progress request	Validation prevents update	Critical
CL-039	Invalid JWT	HTTP 401	High
CL-040	Expired session	Redirect to login	High
CL-041	1000 students using Continue Learning	System remains responsive	High
CL-042	Dashboard loads continue card	Response within acceptable time	Medium
CL-043	Database lookup	Efficient query execution	Medium
CL-044	Lesson deleted after progress saved	Appropriate fallback shown	Medium
CL-045	Course deleted after progress saved	Continue card removed	Medium
CL-046	Internet disconnect while updating progress	Retry or graceful failure	High

Test ID	Test Scenario	Expected Result	Priority
CL-047	Browser crashes during learning	Resume from last saved point	High
CL-048	Duplicate API requests	No duplicate records	High
CL-049	Student unenrolled from course	Continue Learning removed	High
CL-050	Last lesson inaccessible due to permissions	Redirect to valid lesson	High

Module: Work History

Test ID	Test Scenario	Expected Result	Priority
WH-001	Open lesson video	Watch history created	High
WH-002	Watch video for 10 seconds	Watch history updated	High
WH-003	Watch video for 5 minutes	Duration saved	High
WH-004	Watch full video	Completed status updated	High
WH-005	Watch same lesson again	Existing record updated	High
WH-006	Pause video	Current position saved	High
WH-007	Resume video	Starts from saved position	High
WH-008	Close browser during playback	Position saved	High
WH-009	Refresh browser	Position retained	High
WH-010	Fast forward video	Position updated correctly	Medium
WH-011	Watch 25% of video	Duration updated	High
WH-012	Watch 50% of video	Duration updated	High
WH-013	Watch 75% of video	Duration updated	High
WH-014	Watch 100% of video	Duration completed	High
WH-015	Replay video	Duration handled correctly	Medium
WH-016	Dashboard Continue Learning	Latest watch history used	High
WH-017	Resume after logout	Correct position restored	High
WH-018	Resume after login on another device	Position synchronized	High
WH-019	Resume after browser restart	Position restored	High
WH-020	Watch history after lesson completion	Next lesson suggested	High
WH-021	View watch history	Correct history displayed	High
WH-022	Sort by latest watched	Correct order	Medium
WH-023	Filter by course	Correct records	Medium
WH-024	Delete watch history	History removed	Medium

Test ID	Test Scenario	Expected Result	Priority
WH-025	Clear all watch history	All records removed	Medium
WH-026	Watch on desktop	History updated	High
WH-027	Continue on mobile	Position synchronized	High
WH-028	Continue on laptop	Position synchronized	High
WH-029	Simultaneous playback on two devices	Latest progress retained	Medium
WH-030	History record created	Record exists in database	High
WH-031	Duplicate watch entry	Existing record updated	High
WH-032	Invalid lesson ID	Validation error	Medium
WH-033	Invalid user ID	Validation error	Medium
WH-034	GET /watch-history	History returned	High
WH-035	POST /watch-history	History created	High
WH-036	PUT /watch-history	Position updated	High
WH-037	DELETE /watch-history	History removed	Medium
WH-038	Student accesses another student's watch history	Access denied	Critical
WH-039	Invalid JWT	HTTP 401	High
WH-040	Expired token	Redirect to login	High
WH-041	Tampered request	Validation prevents update	Critical
WH-042	Student with 1,000 history records	Loads efficiently	Medium
WH-043	Watch history pagination	Correct pagination	Medium
WH-044	Latest history lookup	Fast response	Medium
WH-045	Internet disconnect during playback	Last valid position retained	High
WH-046	Browser crash	Position restored	High
WH-047	Video removed after watch history exists	Appropriate message shown	Medium
WH-048	Course archived	Watch history preserved or handled per business rule	Medium
WH-049	Duplicate progress update requests	Single record maintained	High
WH-050	Database write failure	Error handled gracefully	High

Module: Bookmarks

Test ID	Test Scenario	Expected Result	Priority
BM-001	Bookmark a lesson	Bookmark created successfully	High

Test ID	Test Scenario	Expected Result	Priority
BM-002	Bookmark a video	Bookmark created successfully	High
BM-003	Bookmark a lesson resource	Bookmark created successfully	Medium
BM-004	Bookmark after lesson completion	Bookmark saved	Medium
BM-005	Bookmark while video is playing	Bookmark saved without interruption	Medium
BM-006	Remove bookmark	Bookmark removed	High
BM-007	Remove bookmarked lesson from lesson page	Removed successfully	Medium
BM-008	Remove bookmark from bookmark list	Removed successfully	Medium
BM-009	Remove non-existing bookmark	Proper validation message	Low
BM-010	View all bookmarks	All bookmarks displayed	High
BM-011	Sort bookmarks by latest	Latest first	Medium
BM-012	Sort bookmarks alphabetically	Correct order	Low
BM-013	Filter bookmarks by course	Correct lessons displayed	Medium
BM-014	Filter bookmarks by module	Correct lessons displayed	Medium
BM-015	Search bookmarked lesson	Correct result displayed	Medium
BM-016	Open bookmarked lesson	Lesson opens successfully	High
BM-017	Resume bookmarked lesson	Opens at last watched position	High
BM-018	Open bookmarked video	Video plays correctly	High
BM-019	Open deleted lesson bookmark	Appropriate message displayed	Medium
BM-020	Refresh page	Bookmark retained	High
BM-021	Logout/Login	Bookmark retained	High
BM-022	Open on another device	Bookmark synchronized	High
BM-023	Browser restart	Bookmark retained	High
BM-024	Bookmark saved in database	Record created	High
BM-025	Duplicate bookmark	Duplicate prevented	High
BM-026	Invalid lesson ID	Validation error	Medium
BM-027	Invalid user ID	Validation error	Medium
BM-028	POST /bookmarks	Bookmark created	High
BM-029	GET /bookmarks	Returns bookmarks	High
BM-030	DELETE /bookmarks/:id	Bookmark removed	High
BM-031	GET /bookmarks/course/:id	Returns course bookmarks	Medium
BM-032	Student accesses another student's bookmarks	Access denied	Critical
BM-033	Invalid JWT	HTTP 401	High

Test ID	Test Scenario	Expected Result	Priority
BM-034	Expired session	Redirect to login	High
BM-035	Tampered bookmark request	Validation prevents modification	Critical
BM-036	Student has 500 bookmarks	Loads efficiently	Medium
BM-037	Bookmark search	Results returned quickly	Medium
BM-038	Bookmark pagination	Correct pagination	Medium
BM-039	Bookmark lesson before enrollment	Access denied	High
BM-040	Delete bookmarked lesson	Bookmark handled correctly	Medium
BM-041	Delete bookmarked course	Related bookmarks handled correctly	Medium
BM-042	Internet disconnect while bookmarking	Retry or graceful error	High
BM-043	Simultaneous bookmark requests	Only one bookmark created	High
BM-044	Bookmark after course completion	Bookmark remains available	Medium
BM-045	Restore deleted lesson	Bookmark becomes usable again	Medium

Module: LearningPath

Test ID	Test Scenario	Expected Result	Priority
LPATH-001	Create learning path	Learning path created	High
LPATH-002	Create path without title	Validation error	High
LPATH-003	Add course to learning path	Course added	High
LPATH-004	Add module to learning path	Module added	Medium
LPATH-005	Save learning path	Saved successfully	High
LPATH-006	Assign path to student	Path assigned	High
LPATH-007	Assign path to batch	Batch assigned	High
LPATH-008	Remove learning path	Assignment removed	Medium
LPATH-009	Update learning path	Changes reflected	High
LPATH-010	Student views assigned path	Correct path displayed	High
LPATH-011	Lessons displayed in correct order	Correct sequence	High
LPATH-012	Modules displayed in order	Correct sequence	High
LPATH-013	Move lesson up	Order updated	Medium
LPATH-014	Move lesson down	Order updated	Medium
LPATH-015	Duplicate lesson order	Validation prevents duplicate	Medium
LPATH-016	Complete prerequisite lesson	Next lesson unlocked	High

Test ID	Test Scenario	Expected Result	Priority
LPATH-017	Skip prerequisite	Access denied	High
LPATH-018	Admin bypass prerequisite	Access granted	Medium
LPATH-019	Mentor bypass prerequisite	Based on permissions	Medium
LPATH-020	Remove prerequisite	Lesson unlocked	Medium
LPATH-021	Mandatory lesson completed	Progress updated	High
LPATH-022	Mandatory lesson skipped	Completion blocked	High
LPATH-023	Optional lesson skipped	Course continues	Medium
LPATH-024	Mandatory quiz not completed	Next lesson locked	High
LPATH-025	Mandatory assignment not submitted	Completion blocked	High
LPATH-026	Progress displayed	Correct percentage	High
LPATH-027	Progress updates after lesson completion	Updated correctly	High
LPATH-028	Module completion	Progress recalculated	High
LPATH-029	Course completion	Learning path updated	High
LPATH-030	Multiple courses completed	Overall progress updated	High
LPATH-031	Next lesson button	Opens next lesson	High
LPATH-032	Previous lesson button	Opens previous lesson	Medium
LPATH-033	Jump to unlocked lesson	Allowed	Medium
LPATH-034	Jump to locked lesson	Access denied	High
LPATH-035	Learning path saved	Database record created	High
LPATH-036	Invalid course ID	Validation error	Medium
LPATH-037	Duplicate assignment	Prevented	High
LPATH-038	Delete learning path	Soft delete applied	Medium
LPATH-039	POST /learning-path	Created successfully	High
LPATH-040	GET /learning-path	Returns assigned path	High
LPATH-041	PUT /learning-path	Updated successfully	High
LPATH-042	DELETE /learning-path	Deleted successfully	Medium
LPATH-043	Student creates learning path	Access denied	Critical
LPATH-044	Student edits learning path	Access denied	Critical
LPATH-045	Invalid JWT	HTTP 401	High
LPATH-046	Unauthorized API access	HTTP 403	High
LPATH-047	Learning path with 500 lessons	Loads efficiently	Medium
LPATH-048	Progress calculation	Response within acceptable time	Medium

Test ID	Test Scenario	Expected Result	Priority
LPATH-049	Multiple students using same path	Performance acceptable	Medium
LPATH-050	Course removed from learning path	Progress recalculated	High
LPATH-051	Lesson deleted from path	Navigation updated	High
LPATH-052	Duplicate lesson in path	Validation prevents duplicate	Medium
LPATH-053	Circular prerequisite configuration	Validation blocks configuration	High
LPATH-054	Student unenrolled from course	Learning path updated	High
LPATH-055	Learning path archived	Student informed appropriately	Medium
LPATH-056	Simultaneous edits by two admins	Conflict handled safely	High

Module: course completion

Test ID	Test Scenario	Expected Result	Priority
CC-001	Complete all lessons	Course marked as completed	High
CC-002	Skip one mandatory lesson	Course not completed	High
CC-003	Complete all mandatory lessons	Eligible for completion	High
CC-004	Complete all optional lessons only	Course remains incomplete	Medium
CC-005	Manual completion by Admin	Course marked complete	Medium
CC-006	Manual completion by Mentor	Based on permissions	Medium
CC-007	All lessons completed	Completion triggered	High
CC-008	Last lesson completed	Completion updated instantly	High
CC-009	Lesson marked incomplete	Course completion revoked	High
CC-010	Deleted completed lesson	Progress recalculated	Medium
CC-011	Newly added lesson after completion	Completion recalculated	High
CC-012	All quizzes passed	Completion allowed	High
CC-013	Quiz failed	Completion blocked	High
CC-014	Quiz retaken and passed	Completion allowed	High
CC-015	Quiz not attempted	Completion blocked	High
CC-016	All assignments submitted	Completion allowed	High
CC-017	Assignment pending	Completion blocked	High
CC-018	Assignment rejected	Completion blocked	High
CC-019	Assignment approved	Completion updated	High
CC-020	Late assignment accepted	Completion updated	Medium

Test ID	Test Scenario	Expected Result	Priority
CC-021	Attendance meets requirement	Completion allowed	High
CC-022	Attendance below minimum	Completion blocked	High
CC-023	Attendance corrected by Admin	Completion recalculated	Medium
CC-024	Course completed	Certificate eligible	High
CC-025	Certificate generated	Student notified	High
CC-026	Completion revoked	Certificate eligibility removed	High
CC-027	Certificate already generated	Duplicate generation prevented	High
CC-028	Student dashboard updated	Course shown as completed	High
CC-029	Mentor dashboard updated	Completion reflected	Medium
CC-030	Admin dashboard updated	Completion count updated	Medium
CC-031	Course completed in learning path	Next course unlocked	High
CC-032	Mandatory course incomplete	Next course locked	High
CC-033	Optional course skipped	Learning path continues	Medium
CC-034	Completion saved in database	Record updated	High
CC-035	Completion timestamp stored	Timestamp accurate	Medium
CC-036	Duplicate completion request	Only one completion recorded	High
CC-037	Invalid course ID	Validation error	Medium
CC-038	GET /course-completion	Returns completion status	High
CC-039	POST /course-completion	Completion recorded	High
CC-040	PUT /course-completion	Status updated	High
CC-041	Invalid completion request	HTTP 400	Medium
CC-042	Student marks own course complete manually	Access denied	Critical
CC-043	Student accesses another student's completion	Access denied	Critical
CC-044	Invalid JWT	HTTP 401	High
CC-045	Unauthorized completion API	HTTP 403	High
CC-046	Completion calculation for 1000 students	Performance acceptable	Medium
CC-047	Bulk completion update	Successful	Medium
CC-048	Dashboard completion statistics	Response within acceptable time	Medium
CC-049	Internet disconnect during completion	Retry or graceful recovery	High
CC-050	Course archived after completion	Completion retained	Medium
CC-051	Student unenrolled after completion	Business rule applied	High
CC-052	Course restored after archive	Completion retained	Medium

Test ID	Test Scenario	Expected Result	Priority
CC-053	Duplicate completion events	Single completion record	High
CC-054	Simultaneous completion updates	No duplicate records	High
CC-055	Database failure during completion	Error handled gracefully	High
CC-056	Course completed	Audit log created	Medium
CC-057	Completion revoked	Audit log created	Medium
CC-058	Certificate eligibility changed	Audit log created	Medium

Module: Learning Analytics

Test ID	Test Scenario	Expected Result	Priority
LA-001	View learning dashboard	Dashboard loads successfully	High
LA-002	View total learning hours	Correct hours displayed	High
LA-003	View completed lessons	Correct count displayed	High
LA-004	View completed courses	Correct count displayed	High
LA-005	View active courses	Correct count displayed	Medium
LA-006	Overall learning progress	Correct percentage displayed	High
LA-007	Course-wise progress	Correct progress displayed	High
LA-008	Module-wise progress	Correct progress displayed	High
LA-009	Lesson-wise progress	Correct progress displayed	Medium
LA-010	Progress after lesson completion	Dashboard updated instantly	High
LA-011	Daily learning time	Correct value displayed	Medium
LA-012	Weekly learning time	Correct value displayed	Medium
LA-013	Monthly learning time	Correct value displayed	Medium
LA-014	Total learning hours	Correct calculation	High
LA-015	Learning time after refresh	Data persists	Medium
LA-016	Total videos watched	Correct count	Medium
LA-017	Average watch duration	Correct value	Medium
LA-018	Most watched lesson	Correct lesson displayed	Low
LA-019	Least watched lesson	Correct lesson displayed	Low
LA-020	Resume statistics	Correct resume count	Low
LA-021	Assignments submitted	Correct count	Medium
LA-022	Pending assignments	Correct count	High

Test ID	Test Scenario	Expected Result	Priority
LA-023	Completed assignments	Correct count	Medium
LA-024	Assignment score average	Correct calculation	Medium
LA-025	Quizzes attempted	Correct count	Medium
LA-026	Quizzes passed	Correct count	Medium
LA-027	Quizzes failed	Correct count	Medium
LA-028	Average quiz score	Correct calculation	Medium
LA-029	Daily learning streak	Correct streak displayed	Medium
LA-030	Streak broken	Reset correctly	Medium
LA-031	Consecutive learning days	Correct count	Medium
LA-032	Weekly streak	Correct calculation	Low
LA-033	Student report	Generated successfully	High
LA-034	Mentor report	Generated successfully	Medium
LA-035	Course report	Generated successfully	Medium
LA-036	Batch report	Generated successfully	Medium
LA-037	Export report	Excel/PDF downloaded	Medium
LA-038	Leaderboard displayed	Correct ranking	Low
LA-039	Ranking updates	Latest ranking shown	Low
LA-040	Top learners	Correct users displayed	Low
LA-041	Analytics stored correctly	Database updated	High
LA-042	Invalid student ID	Validation error	Medium
LA-043	Invalid course ID	Validation error	Medium
LA-044	Duplicate analytics update	No duplicate records	High
LA-045	GET /analytics/student	Returns analytics	High
LA-046	GET /analytics/course	Returns course analytics	Medium
LA-047	GET /analytics/mentor	Returns mentor analytics	Medium
LA-048	GET /analytics/dashboard	Returns dashboard data	High
LA-049	Student views another student's analytics	Access denied	Critical
LA-050	Mentor views unauthorized batch analytics	Access denied	High
LA-051	Invalid JWT	HTTP 401	High
LA-052	Unauthorized analytics API	HTTP 403	High
LA-053	Analytics for 1000 students	Response within acceptable time	High
LA-054	Dashboard analytics load	Fast response	High

Test ID	Test Scenario	Expected Result	Priority
LA-055	Report generation	Completes successfully	Medium
LA-056	Student with no activity	Empty analytics shown	Medium
LA-057	Deleted course	Analytics handled correctly	Medium
LA-058	Deleted student	Analytics retained/archived per business rule	Medium
LA-059	Simultaneous analytics updates	No data corruption	High
LA-060	Database unavailable	Graceful error handling	High
LA-061	Report generated	Audit log created	Medium
LA-062	Analytics exported	Audit log created	Medium
LA-063	Dashboard viewed	Activity logged (if enabled)	Low

Module: Video

Test ID	Test Scenario	Expected Result	Priority
VID-001	Upload MP4 video	Upload successful	High
VID-002	Upload MOV video	Upload successful	Medium
VID-003	Upload unsupported format	Validation error	High
VID-004	Upload empty file	Validation error	High
VID-005	Upload video larger than allowed limit	Error displayed	High
VID-006	Upload duplicate video	Allowed or duplicate warning per business rule	Medium
VID-007	Upload multiple videos	All uploads succeed	Medium
VID-008	Upload video with special characters in filename	Upload successful	Low
VID-009	Generate Direct Upload URL	URL generated	High
VID-010	Upload using Direct Upload URL	Upload succeeds	High
VID-011	Expired Upload URL	Upload rejected	High
VID-012	Invalid Upload URL	Error returned	High
VID-013	Upload interrupted	Retry supported or graceful failure	High
VID-014	Upload completed	Status = Processing	High
VID-015	Processing completed	Status = Ready	High
VID-016	Processing failed	Error logged and shown	High
VID-017	Thumbnail generated	Thumbnail available	Medium
VID-018	Processing notification	Admin notified (if applicable)	Low

Test ID	Test Scenario	Expected Result	Priority
VID-019	Student opens video	Video plays	High
VID-020	Pause video	Playback paused	Medium
VID-021	Resume video	Starts from saved position	High
VID-022	Seek forward	Allowed according to policy	Medium
VID-023	Seek backward	Works correctly	Medium
VID-024	Fullscreen	Works correctly	Low
VID-025	Picture-in-picture	Works if enabled	Low
VID-026	High-speed internet	HD quality streams	Medium
VID-027	Slow internet	Lower quality selected automatically	High
VID-028	Network changes during playback	Quality adjusts automatically	High
VID-029	Manual quality change	Video switches correctly	Medium
VID-030	Stop at 5 minutes	Position saved	High
VID-031	Resume after logout	Position restored	High
VID-032	Resume on another device	Position synchronized	High
VID-033	Refresh browser	Position retained	High
VID-034	Enrolled student opens video	Access granted	High
VID-035	Unenrolled student opens video	Access denied	Critical
VID-036	Expired enrollment	Access denied	High
VID-037	Deleted lesson	Video unavailable	Medium
VID-038	Upload subtitles	Subtitles available	Medium
VID-039	Enable subtitles	Displayed correctly	Medium
VID-040	Disable subtitles	Hidden	Low
VID-041	Invalid subtitle file	Validation error	Medium
VID-042	Replace video	New video available	High
VID-043	Delete video	Video removed	High
VID-044	Restore deleted video metadata	Business rules applied	Medium
VID-045	Update video title	Saved successfully	Medium
VID-046	Update video description	Saved successfully	Medium
VID-047	Access video without JWT	HTTP 401	Critical
VID-048	Tampered video URL	Access denied	Critical
VID-049	Direct Cloudflare URL access	Blocked if using signed URLs	Critical
VID-050	Student accesses another course's video	Access denied	Critical

Test ID	Test Scenario	Expected Result	Priority
VID-051	Expired signed URL	Playback blocked	Critical
VID-052	Invalid video ID	HTTP 404	Medium
VID-053	POST /videos/upload	Upload initiated	High
VID-054	GET /videos	Video list returned	High
VID-055	GET /videos/:id	Video details returned	High
VID-056	PUT /videos/:id	Video updated	High
VID-057	DELETE /videos/:id	Video deleted	High
VID-058	GET /videos/play/:id	Playback URL returned	High
VID-059	Video views	Count updated	Medium
VID-060	Watch duration	Duration tracked	Medium
VID-061	Completion rate	Correct percentage	Medium
VID-062	Most watched video	Analytics displayed	Low
VID-063	1000 students streaming simultaneously	Stable playback	High
VID-064	Upload large file	Upload completes	High
VID-065	Video load time	Playback starts quickly	High
VID-066	CDN caching	Content served efficiently	Medium
VID-067	Internet disconnect during upload	Retry/resume handled	High
VID-068	Browser crash during playback	Resume position retained	High
VID-069	Cloudflare API unavailable	Graceful error shown	High
VID-070	Processing timeout	Retry or error logged	High
VID-071	Duplicate upload requests	Single upload processed	High
VID-072	Course deleted with associated videos	Videos handled according to retention policy	High
VID-073	Video uploaded	Audit log created	Medium
VID-074	Video deleted	Audit log created	Medium
VID-075	Video replaced	Audit log created	Medium

Module: IDE

Test ID	Test Scenario	Expected Result	Priority
IDE-001	Open IDE from LMS	IDE loads successfully	High
IDE-002	Open IDE with valid enrollment	Access granted	High
IDE-003	Open IDE without enrollment	Access denied	High

Test ID	Test Scenario	Expected Result	Priority
IDE-004	IDE loads within acceptable time	IDE ready for coding	High
IDE-005	Refresh IDE page	Current session restored	High
IDE-006	IDE loads on mobile/tablet	Responsive layout	Medium
IDE-007	JWT passed from LMS	User authenticated	High
IDE-008	Expired JWT	Redirect to LMS login	High
IDE-009	Invalid JWT	Access denied	Critical
IDE-010	Student opens another student's IDE	Access denied	Critical
IDE-011	Logout from LMS	IDE session terminated	High
IDE-012	Select Java	Editor configured	High
IDE-013	Select Python	Editor configured	High
IDE-014	Select JavaScript	Editor configured	Medium
IDE-015	Select C++	Editor configured	Medium
IDE-016	Change language after writing code	Confirmation shown	Medium
IDE-017	Unsupported language	Validation error	Medium
IDE-018	Type code	Editor responds correctly	High
IDE-019	Copy & Paste	Works correctly	Medium
IDE-020	Undo	Previous change restored	Medium
IDE-021	Redo	Change reapplied	Medium
IDE-022	Syntax highlighting	Correct highlighting	Medium
IDE-023	Auto indentation	Proper formatting	Medium
IDE-024	Line numbers displayed	Visible	Low
IDE-025	Search in editor	Correct results	Medium
IDE-026	Auto-save after typing	Code saved automatically	High
IDE-027	Refresh browser	Code restored	High
IDE-028	Browser crash	Last saved code restored	High
IDE-029	Logout/Login	Code retained	High
IDE-030	Open from another device	Latest saved code loaded	High
IDE-031	Run valid code	Correct output displayed	High
IDE-032	Run syntax error	Compilation error shown	High
IDE-033	Run runtime error	Runtime error displayed	High
IDE-034	Infinite loop	Execution stopped after timeout	High
IDE-035	Large input	Output generated correctly	Medium

Test ID	Test Scenario	Expected Result	Priority
IDE-036	Multiple executions	Results independent	Medium
IDE-037	Console output	Output displayed	High
IDE-038	Console input	Input accepted	High
IDE-039	Multiple inputs	Processed correctly	Medium
IDE-040	Clear console	Console cleared	Low
IDE-041	Run public test cases	Results displayed	High
IDE-042	Run hidden test cases	Results evaluated	High
IDE-043	Partial success	Passed/failed count shown	High
IDE-044	All test cases passed	Success shown	High
IDE-045	All test cases failed	Failure shown	High
IDE-046	Submit solution	Submission saved	High
IDE-047	Multiple submissions	Latest submission recorded (or per policy)	High
IDE-048	Submission after deadline	Business rule applied	High
IDE-049	Submission with compilation error	Submission handled according to policy	Medium
IDE-050	Submission without running code	Accepted or blocked per business rule	Medium
IDE-051	IDE linked to assignment	Assignment loaded	High
IDE-052	Assignment instructions displayed	Correct content shown	High
IDE-053	Submission updates assignment status	Status updated	High
IDE-054	Mentor views submitted code	Code displayed	High
IDE-055	Compile large program	Performance acceptable	High
IDE-056	100 students executing simultaneously	Stable execution	High
IDE-057	1000 students connected	IDE remains responsive	High
IDE-058	Auto-save under heavy load	No data loss	High
IDE-059	Execute malicious OS command	Blocked by sandbox	Critical
IDE-060	File system access	Denied	Critical
IDE-061	Network access from code	Restricted according to policy	Critical
IDE-062	Access another student's code	Access denied	Critical
IDE-063	Invalid JWT	HTTP 401	High
IDE-064	Unauthorized API request	HTTP 403	High
IDE-065	POST /ide/run	Execution successful	High
IDE-066	POST /ide/submit	Submission stored	High
IDE-067	GET /ide/session	Session loaded	High

Test ID	Test Scenario	Expected Result	Priority
IDE-068	GET /ide/languages	Languages returned	Medium
IDE-069	GET /ide/testcases	Test cases returned	High
IDE-070	Invalid execution request	HTTP 400	Medium
IDE-071	Code saved	Database updated	High
IDE-072	Submission stored	Record created	High
IDE-073	Execution history	Logged correctly	Medium
IDE-074	Duplicate submissions	Handled correctly	Medium
IDE-075	Internet disconnect during execution	Graceful error	High
IDE-076	Browser crash during coding	Code recovered	High
IDE-077	VPS restart during session	User informed and session recovered if possible	High
IDE-078	Compiler unavailable	Friendly error shown	High
IDE-079	Execution timeout	Timeout message displayed	High
IDE-080	Submission while execution running	Proper synchronization	High
IDE-081	Very large source code file	Editor remains responsive	Medium
IDE-082	Unicode characters in code/comments	Saved and displayed correctly	Medium
IDE-083	Code executed	Audit log created	Medium
IDE-084	Code submitted	Audit log created	Medium
IDE-085	Compilation failed	Error logged	Medium

Module: Assignments

Test ID	Test Scenario	Expected Result	Priority
ASSIGN-001	Create assignment with valid details	Assignment created successfully	High
ASSIGN-002	Create assignment without title	Validation error	High
ASSIGN-003	Create assignment without due date	Validation error	High
ASSIGN-004	Create assignment without description	Validation error	Medium
ASSIGN-005	Create assignment with attachments	Assignment created	Medium
ASSIGN-006	Save assignment as draft	Draft saved	Medium
ASSIGN-007	Publish assignment	Assignment visible to students	High
ASSIGN-008	Schedule assignment publication	Published at scheduled time	Medium
ASSIGN-009	Edit assignment title	Updated successfully	High
ASSIGN-010	Edit assignment description	Updated successfully	Medium

Test ID	Test Scenario	Expected Result	Priority
ASSIGN-011	Change due date	Due date updated	High
ASSIGN-012	Add/remove attachments	Changes saved	Medium
ASSIGN-013	Unpublish assignment	Hidden from students	Medium
ASSIGN-014	Delete assignment	Soft delete applied	High
ASSIGN-015	Submit assignment before deadline	Submission successful	High
ASSIGN-016	Submit after deadline	Business rule applied	High
ASSIGN-017	Resubmit assignment	Latest submission handled according to policy	High
ASSIGN-018	Submit without required files	Validation error	High
ASSIGN-019	Submit large file	Upload handled correctly	Medium
ASSIGN-020	Submit empty file	Validation error	Medium
ASSIGN-021	Open coding assignment in IDE	IDE loads assignment	High
ASSIGN-022	Submit code from IDE	Submission recorded	High
ASSIGN-023	Auto-save code before submission	Latest code available	High
ASSIGN-024	Hidden test cases executed	Results evaluated	High
ASSIGN-025	Public test cases executed	Results displayed	High
ASSIGN-026	View submitted assignments	All submissions displayed	High
ASSIGN-027	Download student files	File downloaded	Medium
ASSIGN-028	Evaluate assignment	Marks saved	High
ASSIGN-029	Add feedback	Feedback visible to student	High
ASSIGN-030	Update evaluation	Changes saved	Medium
ASSIGN-031	Reject submission	Student notified	Medium
ASSIGN-032	Student views marks	Correct marks displayed	High
ASSIGN-033	Student views feedback	Feedback displayed	High
ASSIGN-034	Assignment contributes to overall score	Grade updated	High
ASSIGN-035	Assignment completion updates course progress	Progress updated	High
ASSIGN-036	Assignment published	Student notified	Medium
ASSIGN-037	Assignment due reminder	Reminder sent	Medium
ASSIGN-038	Assignment submitted	Mentor notified	Medium
ASSIGN-039	Assignment evaluated	Student notified	Medium
ASSIGN-040	Search assignment by title	Correct assignment displayed	Medium
ASSIGN-041	Filter by course	Correct assignments displayed	Medium

Test ID	Test Scenario	Expected Result	Priority
ASSIGN-042	Filter by submission status	Correct results	Medium
ASSIGN-043	Sort by due date	Correct order	Low
ASSIGN-044	Assignment stored	Record created	High
ASSIGN-045	Submission stored	Record created	High
ASSIGN-046	Marks updated	Database updated	High
ASSIGN-047	Duplicate submission	Business rule enforced	High
ASSIGN-048	POST /assignments	Assignment created	High
ASSIGN-049	GET /assignments	Assignments returned	High
ASSIGN-050	PUT /assignments/:id	Assignment updated	High
ASSIGN-051	DELETE /assignments/:id	Assignment deleted	High
ASSIGN-052	POST /assignments/:id/submit	Submission successful	High
ASSIGN-053	GET /assignments/:id/submissions	Submissions returned	High
ASSIGN-054	Student views another student's submission	Access denied	Critical
ASSIGN-055	Student modifies marks	Access denied	Critical
ASSIGN-056	Mentor evaluates unassigned course	Access denied	High
ASSIGN-057	Invalid JWT	HTTP 401	High
ASSIGN-058	Unauthorized API request	HTTP 403	High
ASSIGN-059	1000 assignment submissions	System remains stable	High
ASSIGN-060	Bulk evaluation	Performance acceptable	Medium
ASSIGN-061	Large attachment uploads	Uploads complete successfully	Medium
ASSIGN-062	Internet disconnect during submission	Retry or graceful recovery	High
ASSIGN-063	Duplicate submission requests	Only one valid submission stored	High
ASSIGN-064	Assignment deleted after submission	Business rules applied	High
ASSIGN-065	Due date changed after submission	Existing submissions remain valid	Medium
ASSIGN-066	Mentor edits marks while student views results	No data inconsistency	High
ASSIGN-067	Database failure during submission	Error handled gracefully	High
ASSIGN-068	Assignment created	Audit log created	Medium
ASSIGN-069	Assignment submitted	Audit log created	Medium
ASSIGN-070	Assignment evaluated	Audit log created	Medium

Module: Payment

Test ID	Test Scenario	Expected Result	Priority
PAY-001	Create payment order	Order created successfully	High
PAY-002	Create order with valid amount	Correct amount generated	High
PAY-003	Create order with zero amount	Validation error	High
PAY-004	Create order with negative amount	Validation error	High
PAY-005	Duplicate order request	Single order created	High
PAY-006	Successful payment	Payment completed	High
PAY-007	Failed payment	Failure status saved	High
PAY-008	Payment cancelled	Cancelled status saved	Medium
PAY-009	Payment timeout	Timeout handled	High
PAY-010	Retry payment	Payment retried successfully	Medium
PAY-011	Verify payment signature	Signature validated	High
PAY-012	Invalid signature	Payment rejected	Critical
PAY-013	Duplicate webhook	Duplicate ignored	Critical
PAY-014	Missing payment ID	Validation error	High
PAY-015	Missing signature	Validation error	High
PAY-016	Successful payment enrolls student	Enrollment created	High
PAY-017	Failed payment	Enrollment not created	High
PAY-018	Duplicate payment notification	Single enrollment	High
PAY-019	Payment pending	Enrollment pending	Medium
PAY-020	Refund issued	Enrollment updated per policy	Medium
PAY-021	Invoice generated	PDF generated	High
PAY-022	Download invoice	Invoice downloaded	High
PAY-023	Invoice contains correct details	Data accurate	High
PAY-024	Duplicate invoice request	Existing invoice returned	Medium
PAY-025	View payment history	All payments displayed	High
PAY-026	Filter payment history	Correct records shown	Medium
PAY-027	Search payment by transaction ID	Correct payment displayed	Medium
PAY-028	Sort payment history	Correct order	Low
PAY-029	Initiate refund	Refund created	High
PAY-030	Partial refund	Correct amount refunded	Medium

Test ID	Test Scenario	Expected Result	Priority
PAY-031	Full refund	Payment fully refunded	Medium
PAY-032	Refund already refunded payment	Validation error	High
PAY-033	Payment success email	Email sent	Medium
PAY-034	Payment failure email	Email sent	Medium
PAY-035	Payment success notification	Dashboard updated	Medium
PAY-036	Invoice email	Delivered successfully	Medium
PAY-037	POST /payments/order	Order created	High
PAY-038	POST /payments/verify	Verification successful	High
PAY-039	GET /payments/history	Returns history	High
PAY-040	POST /payments/refund	Refund processed	High
PAY-041	Tampered payment amount	Payment rejected	Critical
PAY-042	Fake webhook	Rejected	Critical
PAY-043	Invalid JWT	HTTP 401	High
PAY-044	Student accesses another student's payments	Access denied	Critical
PAY-045	1000 simultaneous payment requests	System remains stable	High
PAY-046	Webhook burst handling	All webhooks processed correctly	High
PAY-047	Payment history for large datasets	Fast response	Medium

Module: Certificates

Test ID	Test Scenario	Expected Result	Priority
CERT-001	Student completes course	Certificate generated automatically	High
CERT-002	Admin manually generates certificate	Certificate created	High
CERT-003	Mentor generates certificate	Based on permissions	Medium
CERT-004	Duplicate generation request	Duplicate prevented	High
CERT-005	Course not completed	Certificate not generated	High
CERT-006	Student not enrolled	Certificate not generated	High
CERT-007	Load default template	Template loaded	High
CERT-008	Apply custom template	Template updated	Medium
CERT-009	Update logo	Certificate reflects new logo	Medium
CERT-010	Update authorized signature	Updated correctly	Medium
CERT-011	Change certificate title	Display updated	Low

Test ID	Test Scenario	Expected Result	Priority
CERT-012	Template preview	Preview generated	Medium
CERT-013	Generate PDF	PDF generated successfully	High
CERT-014	Download certificate	PDF downloaded	High
CERT-015	Print certificate	Print works correctly	Medium
CERT-016	PDF opens on mobile	Displayed correctly	Medium
CERT-017	PDF contains correct details	Data verified	High
CERT-018	QR code generated	QR visible	High
CERT-019	Scan QR	Verification page opens	High
CERT-020	Valid QR	Certificate verified	High
CERT-021	Invalid QR	"Certificate Not Found" shown	High
CERT-022	Expired/Revoked certificate QR	Appropriate status shown	High
CERT-023	Search by Certificate ID	Certificate displayed	High
CERT-024	Search by Student Name	Correct certificate displayed (if supported)	Medium
CERT-025	Search invalid Certificate ID	Error displayed	High
CERT-026	Verification page responsive	Works on all devices	Medium
CERT-027	View all certificates	List displayed	High
CERT-028	Filter by course	Correct certificates shown	Medium
CERT-029	Filter by student	Correct certificates shown	Medium
CERT-030	Reissue certificate	New version generated	High
CERT-031	Revoke certificate	Status updated	High
CERT-032	Restore revoked certificate	Status restored	Medium
CERT-033	View certificates	Certificates listed	High
CERT-034	Download certificate	Download successful	High
CERT-035	Share certificate	Share link generated	Medium
CERT-036	Certificate visible after refresh	Data persists	Medium
CERT-037	View certificate statistics	Correct data displayed	Medium
CERT-038	Export certificate list	Excel/PDF generated	Medium
CERT-039	Search certificate	Correct result	Medium
CERT-040	Bulk certificate generation	All certificates generated	High
CERT-041	POST /certificates/generate	Certificate created	High
CERT-042	GET /certificates	Certificate list returned	High
CERT-043	GET /certificates/:id	Certificate details returned	High

Test ID	Test Scenario	Expected Result	Priority
CERT-044	GET /verify/:certificateId	Verification successful	High
CERT-045	DELETE /certificates/:id	Certificate revoked	High
CERT-046	Certificate stored	Record created	High
CERT-047	Unique certificate ID	No duplicates	High
CERT-048	QR token stored	Correct value	Medium
CERT-049	Revocation status updated	Database updated	High
CERT-050	Student downloads another student's certificate	Access denied	Critical
CERT-051	Invalid JWT	HTTP 401	High
CERT-052	Tampered Certificate ID	Validation error	High
CERT-053	Direct PDF URL access without authorization	Access denied or signed URL required	Critical
CERT-054	Generate 1000 certificates	Performance acceptable	High
CERT-055	Bulk download requests	Stable performance	Medium
CERT-056	Verification portal under heavy load	Responsive	Medium
CERT-057	Certificate generated twice	Only one active certificate unless versioning enabled	High
CERT-058	Student name updated after generation	Business rule applied (reissue or retain original)	Medium
CERT-059	Course deleted after certificate issued	Certificate remains verifiable	High
CERT-060	Organization logo changed	Existing certificates follow business rule (unchanged or regenerated)	Medium
CERT-061	Verification service unavailable	Friendly error displayed	High
CERT-062	Database unavailable during generation	Error handled and retry supported	High
CERT-063	Certificate generated	Audit log created	Medium
CERT-064	Certificate revoked	Audit log created	Medium
CERT-065	Certificate verified	Verification event logged (optional)	Low

Module: Placement

Test ID	Test Scenario	Expected Result	Priority
PLC-001	Create company	Company created successfully	High
PLC-002	Edit company details	Changes saved	High
PLC-003	Delete company	Soft delete applied	Medium

Test ID	Test Scenario	Expected Result	Priority
PLC-004	Upload company logo	Logo uploaded	Medium
PLC-005	Search company	Company displayed	Medium
PLC-006	Duplicate company	Validation error	Medium
PLC-007	Create job posting	Job created	High
PLC-008	Edit job posting	Job updated	High
PLC-009	Close job posting	Job unavailable	Medium
PLC-010	Delete job	Soft delete	Medium
PLC-011	Job expiry	Automatically expired	High
PLC-012	Duplicate job	Validation error	Medium
PLC-013	Eligible student	Can apply	High
PLC-014	Ineligible student	Cannot apply	High
PLC-015	Attendance criteria met	Eligible	High
PLC-016	Attendance below minimum	Blocked	High
PLC-017	Course incomplete	Blocked	High
PLC-018	Placement status already placed	Cannot apply	High
PLC-019	Apply for job	Application submitted	High
PLC-020	Apply twice	Duplicate prevented	High
PLC-021	Cancel application	Application cancelled	Medium
PLC-022	View application status	Correct status displayed	High
PLC-023	Apply after deadline	Blocked	High
PLC-024	Resume missing	Validation error	High
PLC-025	Upload resume	Resume uploaded	High
PLC-026	Replace resume	Updated successfully	High
PLC-027	Download resume	Resume downloaded	Medium
PLC-028	Invalid file format	Validation error	Medium
PLC-029	Large resume upload	Upload successful	Medium
PLC-030	Recruiter views applicants	Applicant list displayed	High
PLC-031	Recruiter downloads resume	Resume downloaded	High
PLC-032	Recruiter filters candidates	Correct results	Medium
PLC-033	Recruiter updates status	Status updated	High
PLC-034	Recruiter unauthorized access	Access denied	Critical
PLC-035	Schedule interview	Interview scheduled	High

Test ID	Test Scenario	Expected Result	Priority
PLC-036	Reschedule interview	Schedule updated	Medium
PLC-037	Cancel interview	Interview cancelled	Medium
PLC-038	Student notified	Notification received	High
PLC-039	Mentor notified	Notification received	Medium
PLC-040	Add interview feedback	Feedback saved	Medium
PLC-041	Upload offer letter	Offer saved	High
PLC-042	Student views offer	Offer displayed	High
PLC-043	Student accepts offer	Status updated	High
PLC-044	Student rejects offer	Status updated	Medium
PLC-045	Offer withdrawn	Student notified	Medium
PLC-046	View placement dashboard	Dashboard loads	High
PLC-047	Placement statistics	Correct values displayed	Medium
PLC-048	Active jobs	Correct jobs shown	Medium
PLC-049	Recently placed students	Correct list displayed	Medium
PLC-050	Upcoming interviews	Correct schedule shown	Medium
PLC-051	Placement percentage	Correct calculation	Medium
PLC-052	Company-wise placements	Correct report	Medium
PLC-053	Course-wise placements	Correct report	Medium
PLC-054	Highest package	Correct value	Medium
PLC-055	Average package	Correct value	Medium
PLC-056	Job published	Eligible students notified	High
PLC-057	Interview scheduled	Student notified	High
PLC-058	Offer uploaded	Student notified	High
PLC-059	Job expired	Notification shown	Medium
PLC-060	POST /jobs	Job created	High
PLC-061	GET /jobs	Jobs returned	High
PLC-062	POST /applications	Application created	High
PLC-063	GET /applications	Applications returned	High
PLC-064	POST /interviews	Interview scheduled	High
PLC-065	POST /offers	Offer uploaded	High
PLC-066	Job stored	Record created	High
PLC-067	Application stored	Record created	High

Test ID	Test Scenario	Expected Result	Priority
PLC-068	Offer stored	Record created	High
PLC-069	Duplicate application	Prevented	High
PLC-070	Student accesses another student's application	Access denied	Critical
PLC-071	Student modifies placement status	Access denied	Critical
PLC-072	Invalid JWT	HTTP 401	High
PLC-073	Unauthorized recruiter access	HTTP 403	High
PLC-074	1000 students apply simultaneously	Stable performance	High
PLC-075	Large placement reports	Generated successfully	Medium
PLC-076	Bulk notifications	Delivered successfully	Medium
PLC-077	Job deleted after application	Existing applications handled per business rule	High
PLC-078	Company deleted	Jobs and applications handled correctly	High
PLC-079	Student placed in multiple companies	Business rule enforced	High
PLC-080	Offer uploaded twice	Duplicate prevented	High
PLC-081	Internet disconnect during application	Retry/graceful recovery	High
PLC-082	Database unavailable	Error handled gracefully	High
PLC-083	Job created	Audit log created	Medium
PLC-084	Student applied	Audit log created	Medium
PLC-085	Offer uploaded	Audit log created	Medium

Module: ZOHO

Test ID	Test Scenario	Expected Result	Priority
ZOHO-001	Admission approved	Zoho email account created	High
ZOHO-002	Create email with valid details	Email created successfully	High
ZOHO-003	Duplicate email	Validation error	High
ZOHO-004	Invalid username	Validation error	High
ZOHO-005	Create multiple student accounts	All accounts created	High
ZOHO-006	Send welcome email	Student receives credentials	High
ZOHO-007	Temporary password generated	Password created securely	High
ZOHO-008	Login using Zoho credentials	Login successful	High
ZOHO-009	Force password change on first login	Prompt displayed	High

Test ID	Test Scenario	Expected Result	Priority
ZOHO-010	Disable student	Zoho account disabled	High
ZOHO-011	Reactivate student	Zoho account reactivated	High
ZOHO-012	Delete student	Zoho account handled per retention policy	High
ZOHO-013	Rename student	Email alias/name updated if supported	Medium
ZOHO-014	Reset password	Password reset successfully	High
ZOHO-015	Student changes password	Password updated	High
ZOHO-016	Invalid password policy	Validation error	Medium
ZOHO-017	Password reset notification	Student notified	Medium
ZOHO-018	Student details updated in LMS	Zoho synchronized	High
ZOHO-019	Email already exists	Synchronization handled gracefully	High
ZOHO-020	Failed synchronization	Retry or error logged	High
ZOHO-021	Bulk student import	All accounts synchronized	Medium
ZOHO-022	Send admission email	Delivered successfully	High
ZOHO-023	Send assignment notification	Delivered	Medium
ZOHO-024	Send course completion email	Delivered	Medium
ZOHO-025	Send certificate email	Delivered	Medium
ZOHO-026	Send placement notification	Delivered	Medium
ZOHO-027	View student email status	Correct status displayed	Medium
ZOHO-028	Resend credentials	Email resent	Medium
ZOHO-029	Bulk create accounts	All accounts created	High
ZOHO-030	Bulk disable accounts	All accounts disabled	High
ZOHO-031	POST /zoho/create-user	User created	High
ZOHO-032	POST /zoho/reset-password	Password reset	High
ZOHO-033	POST /zoho/disable-user	User disabled	High
ZOHO-034	GET /zoho/user-status	Status returned	Medium
ZOHO-035	POST /zoho/send-email	Email sent	High
ZOHO-036	Email stored in Users table	Value saved	High
ZOHO-037	Zoho user ID stored	Correct mapping	High
ZOHO-038	Sync status updated	Database reflects current state	Medium
ZOHO-039	Student resets another user's password	Access denied	Critical
ZOHO-040	Invalid Zoho API credentials	Request rejected	High
ZOHO-041	Expired OAuth/API token	Refresh or error handled	High

Test ID	Test Scenario	Expected Result	Priority
ZOHO-042	Unauthorized API request	HTTP 403	High
ZOHO-043	Create 500 student accounts	Performance acceptable	High
ZOHO-044	Bulk email sending	All emails processed	Medium
ZOHO-045	Sync 1000 users	Completed successfully	Medium
ZOHO-046	Zoho service unavailable	Retry/friendly error	High
ZOHO-047	Duplicate account request	Single account created	High
ZOHO-048	Network interruption during provisioning	Safe retry	High
ZOHO-049	Student deleted during sync	Operation handled correctly	High
ZOHO-050	Invalid domain configuration	Configuration error displayed	High
ZOHO-051	Zoho account created	Audit log created	Medium
ZOHO-052	Password reset	Audit log created	Medium
ZOHO-053	Account disabled	Audit log created	Medium

Module: Notification

Test ID	Test Scenario	Expected Result	Priority
NOTIF-001	New notification received	Notification displayed	High
NOTIF-002	Notification badge count	Count updated	High
NOTIF-003	Open notification	Marked as read	High
NOTIF-004	Delete notification	Removed successfully	Medium
NOTIF-005	Mark all as read	All notifications updated	Medium
NOTIF-006	Notification order	Latest appears first	Medium
NOTIF-007	Admission email	Delivered successfully	High
NOTIF-008	Enrollment email	Delivered	High
NOTIF-009	Assignment notification	Delivered	Medium
NOTIF-010	Certificate email	Delivered	High
NOTIF-011	Placement email	Delivered	High
NOTIF-012	Payment receipt email	Delivered	High
NOTIF-013	Admission confirmation	WhatsApp delivered	High
NOTIF-014	Assignment reminder	Delivered	Medium
NOTIF-015	Interview notification	Delivered	High
NOTIF-016	Offer notification	Delivered	High

Test ID	Test Scenario	Expected Result	Priority
NOTIF-017	Certificate notification	Delivered	Medium
NOTIF-018	Create announcement	Saved successfully	High
NOTIF-019	Publish announcement	Visible to target users	High
NOTIF-020	Edit announcement	Changes reflected	Medium
NOTIF-021	Delete announcement	Removed	Medium
NOTIF-022	Schedule announcement	Published at scheduled time	High
NOTIF-023	Target specific batch	Only selected batch receives it	High
NOTIF-024	Enable email notifications	Emails received	Medium
NOTIF-025	Disable email notifications	Emails stopped (except mandatory/system notifications)	Medium
NOTIF-026	Enable WhatsApp	WhatsApp received	Medium
NOTIF-027	Disable WhatsApp	WhatsApp stopped	Medium
NOTIF-028	Enable in-app notifications	In-app notifications shown	Medium
NOTIF-029	Assignment reminder	Sent on schedule	High
NOTIF-030	Interview reminder	Sent on schedule	High
NOTIF-031	Course reminder	Sent on schedule	Medium
NOTIF-032	Payment reminder	Sent on schedule	Medium
NOTIF-033	Notify all students	All receive notification	High
NOTIF-034	Notify mentors	Mentors receive notification	High
NOTIF-035	Notify specific batch	Only selected batch notified	High
NOTIF-036	Notify selected users	Only selected users notified	Medium
NOTIF-037	View notification history	Correct history displayed	Medium
NOTIF-038	Search notifications	Correct results	Medium
NOTIF-039	Filter notifications	Correct records	Medium
NOTIF-040	Export notification history	Export generated	Low
NOTIF-041	POST /notifications	Notification created	High
NOTIF-042	GET /notifications	Notification list returned	High
NOTIF-043	PUT /notifications/read	Status updated	High
NOTIF-044	DELETE /notifications/:id	Notification deleted	Medium
NOTIF-045	POST /notifications/bulk	Bulk notification sent	High
NOTIF-046	Notification stored	Record created	High
NOTIF-047	Read status updated	Database updated	High

Test ID	Test Scenario	Expected Result	Priority
NOTIF-048	Duplicate notification	Prevented if business rule applies	Medium
NOTIF-049	Student reads another student's notification	Access denied	Critical
NOTIF-050	Invalid JWT	HTTP 401	High
NOTIF-051	Unauthorized notification API	HTTP 403	High
NOTIF-052	Tampered notification ID	Validation error	High
NOTIF-053	Send 10,000 notifications	Queue processes successfully	High
NOTIF-054	Load notification center	Fast response	Medium
NOTIF-055	Bulk announcement delivery	Stable performance	High
NOTIF-056	Email provider unavailable	Notification queued or retried	High
NOTIF-057	WhatsApp API unavailable	Retry or graceful failure	High
NOTIF-058	Duplicate notification trigger	Single notification delivered	High
NOTIF-059	User deleted before delivery	Notification discarded or logged	Medium
NOTIF-060	Database unavailable	Queue retained and retried	High
NOTIF-061	Scheduled notification after course deletion	Business rule applied	Medium
NOTIF-062	Notification sent	Audit log created	Medium
NOTIF-063	Bulk announcement published	Audit log created	Medium
NOTIF-064	Notification deleted	Audit log created	Medium

Module: Security

Test ID	Test Scenario	Expected Result	Priority
SEC-001	Login with valid credentials	Login successful	Critical
SEC-002	Invalid password	Access denied	Critical
SEC-003	Invalid email	Access denied	Critical
SEC-004	Empty credentials	Validation error	High
SEC-005	SQL/NoSQL injection in login	Blocked	Critical
SEC-006	Brute-force login attempts	Account temporarily locked or rate limited	Critical
SEC-007	Multiple failed logins	Logged for monitoring	High
SEC-008	Expired password policy	Password reset required	Medium
SEC-009	Valid JWT	Access granted	Critical
SEC-010	Invalid JWT	HTTP 401	Critical

Test ID	Test Scenario	Expected Result	Priority
SEC-011	Expired JWT	HTTP 401	Critical
SEC-012	Tampered JWT	Access denied	Critical
SEC-013	JWT with wrong signature	Rejected	Critical
SEC-014	Refresh token reuse	Old refresh token rejected	Critical
SEC-015	Logout invalidates refresh token	Session cannot be reused	High
SEC-016	Student accesses Admin API	HTTP 403	Critical
SEC-017	Mentor accesses Admin settings	HTTP 403	Critical
SEC-018	Student accesses Mentor dashboard	HTTP 403	Critical
SEC-019	Admin accesses all modules	Allowed	High
SEC-020	Unauthorized role modification	Blocked	Critical
SEC-021	Logout	Session invalidated	High
SEC-022	Logout from all devices	All sessions invalidated	High
SEC-023	Session timeout after inactivity	Re-authentication required	High
SEC-024	Concurrent sessions (same user)	Handled according to policy	Medium
SEC-025	Stolen session token reuse	Rejected if invalidated	Critical
SEC-026	Missing Authorization header	HTTP 401	Critical
SEC-027	Invalid HTTP method	HTTP 405	Medium
SEC-028	Missing required fields	HTTP 400	High
SEC-029	Oversized request body	Rejected	High
SEC-030	API rate limiting	Excess requests blocked	Critical
SEC-031	Inject {"\$ne": null} into login	Blocked	Critical
SEC-032	Inject \$gt operator	Blocked	Critical
SEC-033	Inject \$regex	Blocked	Critical
SEC-034	Inject \$where	Blocked	Critical
SEC-035	Malicious JSON payload	Validation error	Critical
SEC-036	<script>alert(1)</script> in profile	Escaped or rejected	Critical
SEC-037	XSS in announcements	Blocked	Critical
SEC-038	XSS in assignment submission	Blocked	Critical
SEC-039	XSS in forum/comments (future)	Blocked	Critical
SEC-040	Stored XSS attempt	Sanitized	Critical
SEC-041	POST without CSRF protection (if cookie auth)	Blocked	High

Test ID	Test Scenario	Expected Result	Priority
SEC-042	Forged payment request	Blocked	Critical
SEC-043	Forged profile update	Blocked	High
SEC-044	Upload executable (.exe)	Rejected	Critical
SEC-045	Upload JavaScript file	Rejected	Critical
SEC-046	Upload oversized file	Rejected	High
SEC-047	Upload renamed executable (.exe → .pdf)	MIME validation fails	Critical
SEC-048	Upload malware test file	Quarantined/rejected	Critical
SEC-049	Direct Cloudflare URL access	Blocked if signed URLs are used	Critical
SEC-050	Share video URL with another student	Unauthorized user denied	Critical
SEC-051	Expired signed URL	Playback blocked	Critical
SEC-052	Access unpublished video	Denied	High
SEC-053	Execute OS commands	Blocked	Critical
SEC-054	File system access	Blocked	Critical
SEC-055	Network access	Restricted per policy	Critical
SEC-056	Infinite loop	Execution timeout enforced	Critical
SEC-057	Excessive memory usage	Process terminated	Critical
SEC-058	Fake Razorpay webhook	Rejected	Critical
SEC-059	Invalid Razorpay signature	Rejected	Critical
SEC-060	Payment amount tampering	Rejected	Critical
SEC-061	Replay webhook	Duplicate ignored	Critical
SEC-062	Access soft-deleted data	Hidden unless authorized	High
SEC-063	Unauthorized direct database access	Blocked	Critical
SEC-064	Sensitive fields excluded from APIs	Password hashes/tokens never exposed	Critical
SEC-065	Audit log tampering attempt	Prevented	Critical
SEC-066	Password stored	Hashed (e.g., bcrypt/Argon2)	Critical
SEC-067	JWT secret protected	Not exposed	Critical
SEC-068	HTTPS enforced	Secure transport only	Critical
SEC-069	Sensitive environment variables	Not exposed to client	Critical
SEC-070	Failed login	Logged	High
SEC-071	Admin deletes course	Audit log created	High
SEC-072	Role updated	Audit log created	High
SEC-073	Payment verified	Audit log created	High

Test ID	Test Scenario	Expected Result	Priority
SEC-074	10,000 login attempts	Rate limiting enforced	Critical
SEC-075	API flood	Service remains available	High
SEC-076	Large invalid payloads	Rejected safely	High
SEC-077	Token expires during API call	Graceful 401 response	High
SEC-078	Server restart	Active sessions handled correctly	Medium
SEC-079	Clock skew affects JWT expiry	Small tolerance handled	Medium
SEC-080	Third-party security service unavailable	Fail safely with logging	High

Module: Performance

Test ID	Test Scenario	Expected Result	Priority
PERF-001	100 users login simultaneously	All logins succeed	High
PERF-002	500 users login simultaneously	Response time acceptable	High
PERF-003	1000 users login simultaneously	No server crash	Critical
PERF-004	Failed login flood	Rate limiting triggered	High
PERF-005	Peak-hour login (9 AM)	Stable performance	High
PERF-006	Student dashboard load	<2 seconds	High
PERF-007	Mentor dashboard load	<2 seconds	High
PERF-008	Admin dashboard load	<3 seconds	High
PERF-009	Dashboard refresh	Fast response	Medium
PERF-010	Dashboard with 1000 records	Responsive	High
PERF-011	1000 concurrent reads	Stable	High
PERF-012	500 concurrent writes	Stable	High
PERF-013	Bulk insert	Completed successfully	Medium
PERF-014	Complex aggregation query	Response within acceptable time	Medium
PERF-015	Database indexing effectiveness	Query uses indexes	High
PERF-016	Load 500 courses	Fast response	Medium
PERF-017	Search courses	<1 second	High
PERF-018	Course filtering	Fast response	Medium
PERF-019	Pagination	Correct and efficient	High
PERF-020	100 students stream simultaneously	Smooth playback	High
PERF-021	500 concurrent streams	No buffering from origin	High

Test ID	Test Scenario	Expected Result	Priority
PERF-022	1000 concurrent streams	Stable CDN delivery	Critical
PERF-023	Adaptive quality switching	Seamless	Medium
PERF-024	Resume playback	Immediate resume	Medium
PERF-025	100 code executions	Stable	High
PERF-026	500 concurrent executions	Stable	High
PERF-027	1000 IDE users connected	No crashes	Critical
PERF-028	Large code compilation	Acceptable execution time	Medium
PERF-029	Auto-save under load	No data loss	High
PERF-030	1000 assignment submissions	Successful	High
PERF-031	Bulk file uploads	Stable	Medium
PERF-032	Mentor reviews many submissions	Responsive	Medium
PERF-033	500 payment requests	Stable	High
PERF-034	Razorpay webhook burst	All webhooks processed	High
PERF-035	Payment history query	Fast response	Medium
PERF-036	Send 10,000 emails	Queue processes successfully	High
PERF-037	Send 5,000 WhatsApp messages	Queue processes successfully	High
PERF-038	Bulk in-app notifications	Delivered successfully	Medium
PERF-039	GET APIs under load	<500 ms average	High
PERF-040	POST APIs under load	<1 second average	High
PERF-041	PUT APIs under load	Stable	Medium
PERF-042	DELETE APIs under load	Stable	Medium
PERF-043	API rate limiting	Activated correctly	High
PERF-044	Upload 1 GB recording	Upload completes	High
PERF-045	Upload multiple videos	All uploads succeed	Medium
PERF-046	Generate certificates in bulk	Successful	Medium
PERF-047	CPU utilization	Within acceptable threshold	High
PERF-048	Memory utilization	Stable without leaks	High
PERF-049	Disk usage	No storage bottlenecks	Medium
PERF-050	Network bandwidth	No saturation	High
PERF-051	1500 concurrent users	System degrades gracefully	High
PERF-052	2000 concurrent users	No data corruption	High
PERF-053	Database under heavy load	No crashes	High

Test ID	Test Scenario	Expected Result	Priority
PERF-054	API overload	Queue or throttling applied	High
PERF-055	Run continuously for 24 hours	Stable	High
PERF-056	Continuous video streaming	Stable	High
PERF-057	Continuous IDE execution	Stable	High
PERF-058	Continuous notifications	Stable	Medium
PERF-059	Restart backend during load	Service recovers	High
PERF-060	Restart MongoDB	Application reconnects	High
PERF-061	VPS reboot	Services start automatically	High
PERF-062	Internet interruption	Recovery handled gracefully	Medium
PERF-063	CPU alerts	Alert generated	Medium
PERF-064	RAM alerts	Alert generated	Medium
PERF-065	Disk alerts	Alert generated	Medium
PERF-066	Slow API logging	Logged correctly	High
PERF-067	Failed job logging	Logged correctly	High

Module: API Testing

Test ID	API	Test Scenario	Expected Result	Priority
API-001	POST /auth/login	Valid login	HTTP 200 + JWT	Critical
API-002	POST /auth/login	Invalid password	HTTP 401	Critical
API-003	POST /auth/login	Missing email	HTTP 400	High
API-004	POST /auth/refresh	Valid refresh token	New JWT returned	High
API-005	POST /auth/logout	Logout	Refresh token invalidated	High
API-006	GET /users	Fetch users	List returned	High
API-007	GET /users/:id	Valid user	User details	High
API-008	POST /users	Create user	User created	High
API-009	PUT /users/:id	Update user	Updated successfully	High
API-010	DELETE /users/:id	Soft delete	User archived	High
API-011	GET /courses	Fetch courses	List returned	High
API-012	GET /courses/:id	Valid course	Course returned	High
API-013	POST /courses	Create course	Course created	High
API-014	PUT /courses/:id	Update course	Updated	High

Test ID	API	Test Scenario	Expected Result	Priority
API-015	DELETE /courses/:id	Delete course	Soft delete	High
API-016	POST /enrollments	Enroll student	Enrollment created	High
API-017	GET /enrollments	Fetch enrollments	List returned	High
API-018	DELETE /enrollments/:id	Cancel enrollment	Status updated	Medium
API-019	GET /progress	Fetch progress	Progress returned	High
API-020	POST /progress	Save progress	Updated	High
API-021	PUT /progress	Update progress	Updated	High
API-022	GET /continue-learning	Continue learning	Correct lesson	High
API-023	POST /videos/upload	Upload video	Upload initiated	High
API-024	GET /videos	Fetch videos	List returned	High
API-025	GET /videos/:id	Fetch video	Video details	High
API-026	DELETE /videos/:id	Delete video	Soft delete	Medium
API-027	POST /ide/run	Run code	Output returned	Critical
API-028	POST /ide/submit	Submit solution	Stored	High
API-029	GET /ide/session	Resume session	Session loaded	High
API-030	POST /assignments	Create assignment	Created	High
API-031	POST /assignments/:id/submit	Submit assignment	Stored	High
API-032	GET /assignments	Fetch assignments	Returned	High
API-033	POST /payments/order	Create order	Order created	High
API-034	POST /payments/verify	Verify payment	Verified	High
API-035	GET /payments/history	Fetch payments	Returned	Medium
API-036	POST /certificates/generate	Generate certificate	Generated	High
API-037	GET /certificates	Fetch certificates	Returned	High
API-038	GET /verify/:certificateId	Verify certificate	Verified	High
API-039	GET /jobs	Fetch jobs	Returned	High
API-040	POST /applications	Apply for job	Application created	High
API-041	GET /applications	Fetch applications	Returned	High
API-042	GET /notifications	Fetch notifications	Returned	High
API-043	PUT /notifications/read	Mark as read	Updated	Medium
API-044	POST /notifications/bulk	Bulk send	Notifications queued	High
API-045	Invalid HTTP method	HTTP 405	High	
API-046	Missing Authorization header	HTTP 401	Critical	

Test ID	API	Test Scenario	Expected Result	Priority
API-047	Invalid JWT	HTTP 401	Critical	
API-048	Expired JWT	HTTP 401	Critical	
API-049	Invalid ObjectID	HTTP 400	High	
API-050	Missing required field	HTTP 400	High	
API-051	Invalid request body	HTTP 400	High	
API-052	Unknown endpoint	HTTP 404	Medium	
API-053	Unsupported media type	HTTP 415	Medium	
API-054	Duplicate request	Handled correctly	High	
API-055	Pagination	Correct records returned	Medium	
API-056	Sorting	Correct order	Medium	
API-057	Filtering	Correct results	Medium	
API-058	Search	Matching results	Medium	
API-059	Login rate limit	HTTP 429 after threshold	Critical	
API-060	Public API rate limit	Rate limiting applied	High	
API-061	Burst requests	Service remains stable	High	
API-062	GET response time	<500 ms average	High	
API-063	POST response time	<1 second average	High	
API-064	Concurrent API requests	Stable responses	High	
API-065	Successful request	Logged if applicable	Medium	
API-066	Failed request	Error logged	High	
API-067	Internal server error	Logged with trace ID	High	

Module: Database

Test ID	Test Scenario	Expected Result	Priority
DB-001	Create new user	Record created successfully	Critical
DB-002	Duplicate email	Validation prevents duplicate	Critical
DB-003	Duplicate phone number	Validation prevents duplicate	High
DB-004	Password stored	Password hashed only	Critical
DB-005	Soft delete user	deletedAt populated; record excluded from active queries	High
DB-006	Assign role to user	Mapping created	High
DB-007	Remove role	Mapping removed	Medium

Test ID	Test Scenario	Expected Result	Priority
DB-008	Duplicate role assignment	Prevented	High
DB-009	Invalid permission mapping	Validation error	High
DB-010	Create course	Record created	High
DB-011	Course without category	Validation error	High
DB-012	Duplicate course code	Prevented	High
DB-013	Soft delete course	Hidden from active queries	High
DB-014	Module linked to course	Valid reference	High
DB-015	Lesson linked to module	Valid reference	High
DB-016	Delete module	Lessons handled according to business rule	High
DB-017	Invalid lesson reference	Validation error	High
DB-018	Student enrollment	Record created	High
DB-019	Duplicate enrollment	Prevented	High
DB-020	Delete enrolled course	Business rule enforced	High
DB-021	Save lesson progress	Record updated	High
DB-022	Update progress	Existing record updated	High
DB-023	Invalid lesson reference	Validation error	High
DB-024	Progress percentage	Value within valid range (0–100)	High
DB-025	Store Cloudflare Video ID	Stored correctly	High
DB-026	Duplicate Cloudflare Video ID	Prevented	High
DB-027	Delete lesson	Video handled per retention policy	Medium
DB-028	Save source code	Stored correctly	High
DB-029	Save execution history	Record created	High
DB-030	Save submission	Linked to assignment	High
DB-031	Store assignment	Record created	High
DB-032	Store submission	Record created	High
DB-033	Duplicate submission	Business rule enforced	High
DB-034	Store Razorpay payment	Record created	High
DB-035	Duplicate payment ID	Prevented	Critical
DB-036	Store refund	Linked correctly	High
DB-037	Unique certificate ID	No duplicates	Critical
DB-038	Store QR verification token	Stored correctly	High
DB-039	Revoke certificate	Status updated	High

Test ID	Test Scenario	Expected Result	Priority
DB-040	Store company	Record created	High
DB-041	Store application	Linked to student & job	High
DB-042	Duplicate application	Prevented	High
DB-043	Store notification	Record created	High
DB-044	Update read status	Saved correctly	High
DB-045	Delete notification	Soft delete or archive applied	Medium
DB-046	User → StudentProfile	Reference valid	High
DB-047	Course → Modules	Reference valid	High
DB-048	Module → Lessons	Reference valid	High
DB-049	Student → Enrollment	Reference valid	High
DB-050	Enrollment → Progress	Reference valid	High
DB-051	Assignment → Submission	Reference valid	High
DB-052	Job → Application	Reference valid	High
DB-053	Email index	Query uses index	High
DB-054	User code index	Query uses index	High
DB-055	Course code index	Query uses index	High
DB-056	Enrollment lookup	Indexed query	High
DB-057	Payment ID lookup	Indexed query	High
DB-058	Payment + Enrollment	Both succeed or both rollback	Critical
DB-059	Certificate generation	Record saved atomically	High
DB-060	Bulk student creation	No partial failures	High
DB-061	Assignment submission	Submission and progress stay consistent	High
DB-062	Backup database	Backup successful	High
DB-063	Restore backup	Data restored correctly	High
DB-064	Restore latest snapshot	Latest data recovered	High
DB-065	User updated	Audit record created	High
DB-066	Course updated	Audit record created	High
DB-067	Payment verified	Audit record created	High
DB-068	Certificate generated	Audit record created	High
DB-069	Query 1 million records	Performance acceptable	High
DB-070	Aggregation pipeline	Executes efficiently	High
DB-071	Bulk inserts	Completed successfully	High

Test ID	Test Scenario	Expected Result	Priority
DB-072	Concurrent writes	No corruption	High
DB-073	Password hash not exposed	Sensitive fields excluded	Critical
DB-074	JWT refresh token stored securely	Protected	Critical
DB-075	Deleted records inaccessible	Hidden from normal queries	High
DB-076	Unauthorized DB query	Access denied	Critical
DB-077	Server restart during write	No corrupted records	High
DB-078	Duplicate concurrent inserts	Single valid record	High
DB-079	Disk full	Graceful failure	High
DB-080	Network interruption	Data consistency maintained	High

Module: VPS Testing

Test ID	Test Scenario	Expected Result	Priority
SYS-001	User loses internet during login	Graceful retry message	High
SYS-002	Internet disconnect during active session	Session resumes after reconnect	High
SYS-003	Browser crash	Session recoverable	High
SYS-004	Browser refresh during payment	Payment status synchronized	High
SYS-005	Session timeout while watching video	Re-authentication handled gracefully	High
SYS-006	Backend service restart	Application reconnects automatically	High
SYS-007	API service unavailable	Friendly error message	High
SYS-008	Backend crashes during assignment submission	No duplicate or corrupted data	Critical
SYS-009	Backend crashes during enrollment	Transaction handled safely	Critical
SYS-010	Backend restart during certificate generation	Generation resumes or fails safely	High
SYS-011	MongoDB restart	Backend reconnects	High
SYS-012	Database unavailable	Graceful error	High
SYS-013	Database timeout	Retry or timeout response	High
SYS-014	Write failure	No partial data committed	Critical
SYS-015	Read failure	Proper error returned	High
SYS-016	Cloudflare unavailable	Student informed gracefully	High
SYS-017	Video processing delayed	Status shown correctly	Medium
SYS-018	Upload interrupted	Resume or retry	High

Test ID	Test Scenario	Expected Result	Priority
SYS-019	CDN latency spike	Adaptive streaming continues	Medium
SYS-020	IDE server restart	Session recovered if possible	High
SYS-021	Compilation server unavailable	Friendly error displayed	High
SYS-022	Code execution timeout	Timeout handled	High
SYS-023	Auto-save during restart	Latest saved version retained	High
SYS-024	Razorpay unavailable	Payment cannot start; user informed	High
SYS-025	Webhook delayed	Enrollment after successful verification	High
SYS-026	Duplicate webhook	Single payment processed	Critical
SYS-027	Payment success but DB write fails	Reconciliation process available	Critical
SYS-028	Zoho API unavailable	Request queued or retry initiated	High
SYS-029	Email delivery failure	Retry or status logged	High
SYS-030	Student email not created	Admin notified	High
SYS-031	Assignment upload interrupted	Resume or retry	High
SYS-032	Video upload interrupted	Resume or retry	High
SYS-033	Large file upload timeout	Proper timeout handling	High
SYS-034	Two admins edit same course	Conflict handled	High
SYS-035	Two mentors evaluate same assignment	Conflict handled	High
SYS-036	Student submits assignment twice simultaneously	Single valid submission	High
SYS-037	Two admins generate same certificate	Duplicate prevented	High
SYS-038	VPS power failure	Automatic restart	High
SYS-039	Power failure during payment	Payment reconciliation	Critical
SYS-040	Power failure during DB write	No corruption	Critical
SYS-041	Restore yesterday's backup	Data restored	High
SYS-042	Restore latest snapshot	Successful	High
SYS-043	Verify restored data	Data integrity maintained	High
SYS-044	Backup while users active	Backup completed safely	Medium
SYS-045	Backend health check	Healthy status returned	Medium
SYS-046	IDE health check	Healthy status returned	Medium
SYS-047	Database health check	Healthy status returned	Medium
SYS-048	Cloudflare availability check	Healthy status returned	Low
SYS-049	Disk reaches 95% usage	Alert generated	High

Test ID	Test Scenario	Expected Result	Priority
SYS-050	RAM reaches 90% usage	Alert generated	High
SYS-051	CPU reaches 95% usage	Alert generated	High
SYS-052	Network saturation	Requests throttled if necessary	High
SYS-053	API error	Logged and alert generated	High
SYS-054	Database connection failure	Logged and alert generated	High
SYS-055	Payment failure	Logged	High
SYS-056	Video upload failure	Logged	Medium
SYS-057	IDE execution failure	Logged	Medium
SYS-058	Complete VPS failure	Restore from backup	Critical
SYS-059	MongoDB data corruption	Restore from backup	Critical
SYS-060	Cloudflare account issue	Fallback procedure documented	High
SYS-061	Domain/DNS issue	Recovery procedure followed	High
SYS-062	SSL certificate expiry	Renewal process executed	High
SYS-063	1000 students online	Stable system	Critical
SYS-064	100 mentors active	Stable system	High
SYS-065	10 admins working simultaneously	Stable system	High
SYS-066	Full LMS operational test	All modules functioning	Critical
SYS-067	End-to-end admission → certificate → placement flow	Entire workflow successful	Critical